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A survey of cyber threat attribution: Challenges, techniques, and future directions

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ABSTRACT

The escalating sophistication of cyberattacks, exemplified by supply chain compromises, AI-driven obfuscation, and politically motivated campaigns, makes accurate attribution a critical yet elusive challenge for national security and economic stability. The inability to reliably trace attacks to their source undermines deterrence, distorts policy responses, and erodes trust in digital ecosystems. Traditional methods struggle with the sheer volume of digital evidence, rapidly evolving adversary tactics, and the inherent complexities of cross-border operations. Moreover, existing literature often provides fragmented analyses, focuses narrowly on cyber threat intelligence sharing or specific threat types, or predates significant advancements in AI/ML tailored for attribution. This survey offers a comprehensive, interdisciplinary review of cyber threat attribution, bridging these critical gaps by systematically analyzing its multifaceted dimensions: technical, legal, geopolitical, social, and economic. Employing a rigorous, PRISMA-ScR compliant methodology that included structured screening and quality assessment across six major databases, we critically appraise current techniques and identify a paradigm shift toward data-driven, intelligent approaches. A key contribution is our novel taxonomy, which structures attribution research by attribution confidence & granularity (the Level of attribution), analytical domains (the “How” and “Where” of evidence processing) and adversarial motivation & profile (the “Why” and “Who”), providing a crucial framework for systematic cross-study comparisons in a complex field. Our findings underscore the transformative potential of emerging AI/ML techniques, particularly graph neural networks, in automating analysis, identifying subtle patterns, and extracting crucial insights from vast datasets, thereby revolutionizing attribution accuracy. This research provides actionable insights for practitioners and policymakers, offering a comprehensive roadmap to advance cyber defense and foster a more resilient and secure global digital ecosystem.

1. Introduction

The digital revolution has ushered in unprecedented connectivity and equally unprecedented vulnerabilities. High-profile cyberattacks like SolarWinds and NotPetya exemplify a disturbing trend: adversaries now exploit supply chains, weaponize AI, and blur the lines between criminal and state-sponsored activity (Saeed et al., 2023). While organizations invest heavily in cyber defenses (Fadziso et al., 2023), one critical capability lags behind: attribution. The inability to reliably trace attacks to their source undermines deterrence, distorts policy responses, and erodes trust in digital ecosystems (Kida and Olukoya, 2023; Kijewski et al., 2016).

Cyber threat attribution, the process of identifying attack perpetrators and motives (Warikoo, 2021), is foundational to effective cybersecurity. Beyond enabling targeted countermeasures (Irshad and

Siddiqui, 2023; Egloff, 2020), it exposes adversary tactics, techniques and procedures (TTPs) (Anisetti et al., 2020; Morag and Maouda, 2021) and informs strategic decisions from diplomacy (e.g., sanctions for state-sponsored attacks) to legal prosecution (Egloff, 2020; Shamsi et al., 2016; Tsagourias and Farrell, 2020; Egloff and Smeets, 2023). Yet attribution today faces existential challenges. Adversaries employ AI-driven obfuscation, exploit jurisdictional arbitrage, and manipulate geopolitical narratives to obscure their tracks. Even conclusive attributions (e.g., WannaCry's ties to North Korea) often arrive too late to mitigate damage or spark international consensus (Colbaugh and Glass, 2011; Anisetti et al., 2020; Morag and Maouda, 2021).

While existing research has yielded valuable insights into specific threat types or technical aspects of attribution, a holistic understanding of the field remains elusive. The dynamic nature of cyber threats,

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combined with the limitations of traditional attribution methods, necessitates a comprehensive survey that critically evaluates the current landscape, identifies key challenges and limitations, and proposes future research directions to enhance attribution capabilities in the face of evolving threats (Gupta et al., 2020; Sindiramutty, 2023). To organize this complex landscape, we introduce a novel taxonomy that structures attribution research along three key dimensions: (1) Attribution Confidence & Granularity (The “Level” of Attribution), (2) Analytical Domains (The “How” and “Where” of Evidence Processing), and (3) Adversarial Motivation & Profile (The “Why” and “Who” Classification). This framework addresses the field’s multifaceted nature encompassing technical, social, political, legal, and economic factors (Rid and Buchanan, 2015; Egloff and Smeets, 2023; Swate et al., 2024) that prior surveys have treated in isolation.

1.1. Motivation

The escalating frequency and sophistication of cyberattacks exemplified by incidents like SolarWinds and NotPetya, establish accurate and timely attribution as a cornerstone of cybersecurity. Attribution transcends mere blame; it is fundamental for effective deterrence, enabling proactive defense, and underpinning justice through accountability (Riggs et al., 2023). It critically informs national and international policy decisions, fosters cross-border investigations, and strengthens international cooperation against cybercrime, ultimately shaping the wider landscape of cyber conflict (Wagner et al., 2019; Wasel, 2023; Mishra et al., 2022). By understanding adversary TTPs, defenders can tailor security measures (Sun et al., 2023; Irshad and Siddiqui, 2024), while the prospect of identification acts as a deterrent, though its effect varies across ideologically or politically motivated actors (Baliga et al., 2020; Guitton, 2012).

Yet, achieving robust attribution is fraught with challenges. Adversaries employ sophisticated deception, anonymization (Han et al., 2018; Basak et al., 2019), and exploit geopolitical complexities (Cordey, 2023). This renders attribution inherently probabilistic, requiring analysis of technical evidence alongside broader contexts. Traditional methods, often reliant on static indicators (Bowman and Huang, 2021; Bhardwaj et al., 2022), struggle with the massive volumes of dynamic data generated (Colbaugh and Glass, 2011). While machine learning (ML) offers promising solutions to analyze vast datasets and identify subtle patterns (Rawat et al., 2021; Apruzzese et al., 2023) for enhanced threat detection (Nakhodchi et al., 2021; Bilen and Özer, 2021), its application introduces ethical and legal concerns, including algorithmic bias, transparency deficits, data privacy issues, and adversarial false flag campaigns that hinder accuracy (Egloff and Smeets, 2023; Pahi and Skopik, 2019; Hettema, 2021). Furthermore, incorrect or misattribution carries severe consequences, escalating tensions and potentially harming innocent parties (Hettema, 2021). Even accurate attribution can strain diplomatic relations and be politically weaponized to achieve geopolitical objectives (Schulzke, 2018; Stoddart, 2022), underscoring the need for meticulous investigation and transparency.

Prior surveys, however, have not systematically examined AI/ML’s role in attribution or synthesized its interdisciplinary nature. Most focus narrowly on cyber threat intelligence sharing or specific threat types, often predating recent AI advances and overlooking critical interdependencies. This survey fills this gap by providing a comprehensive, analytical review of attribution research, critically evaluating challenges and solutions. We introduce a novel taxonomy that structures the field by attribution confidence & granularity, analytical domains, and adversarial motivation & profile, equipping researchers and policymakers with a structured framework to advance attribution accuracy and resilience.

1.2. Related surveys and contributions

A deep analysis of the existing literature on cyber threat intelligence (CTI) and cyber threat attribution (CTA) reveals significant and critical gaps. Many CTI surveys overwhelmingly focus on information sharing mechanisms (Table 1), while those that do address attribution (Table 2) often suffer from a narrow focus on specific threats like Advanced Persistent Threats (APTs). This narrowness leads to an “attribution myopia”, overlooking crucial aspects such as complex ransomware ecosystems and sophisticated supply chain compromises. Furthermore, a substantial portion of current methodologies remains rooted in a pre-AI era, significantly lagging behind rapid advancements in AI/ML techniques (Sharma et al., 2023; Tang et al., 2022) and failing to adequately address emerging adversarial innovations like AI-driven false flags.

A critical overarching issue is the pervasive disciplinary silo effect, which results in a fundamental lack of an interdisciplinary, systematic framework for attribution analysis. Technical studies frequently omit considerations for legal admissibility of evidence, while geopolitical analyses often lack the necessary forensic grounding for verifiable attribution. This fragmentation prevents a holistic understanding of attribution’s multifaceted nature, spanning its technical, legal, and geopolitical dimensions, and leaves the field without a unified approach to account for dynamic adversarial innovations. This survey fills these gaps with three key contributions:

Specifically, our contributions include:

- (1) A comprehensive analysis and synthesis of attribution challenges, features, and approaches.
- (2) A novel taxonomy that categorizes attribution techniques granularly along three dimensions, enabling systematic interdisciplinary analysis and insights.
- (3) A roadmap for future research, highlighting urgent needs such as graph neural networks for attribution (Naseer, 2024; Chatzimanetoglou and Rantos, 2024) and standardized evaluation metrics for attribution systems (Yang et al., 2009; Alghamdi, 2021).

1.3. Survey methodologies

To ensure rigorous reproducibility, our survey employed a PRISMA-ScR compliant systematic methodology, beginning with comprehensive searches across six major academic databases (IEEE Xplore, Web of Science, Scopus, ACM DL, SpringerLink, PubMed). Using the Boolean query (“cyber threat attribution” OR “attack attribution”) AND (“framework” OR “technique” OR “AI” OR “machine learning” OR “geopolitical” OR “legal”), we captured foundational works (2000–2025) and recent AI/ML advances (2017–2025), yielding 2318 initial records. Beyond this initial search, we augmented our selection through citation chaining (snowballing), reviewing references of highly relevant papers and identifying key citing articles to ensure broad coverage of influential works. A two-phase screening process applied explicit inclusion criteria (peer-reviewed empirical studies/reviews primarily focused on cyber threat attribution methods, challenges, or dimensions, in English) and detailed exclusion criteria (non-peer-reviewed reports, tangential topics, methodologically unsound studies, duplicates, or inaccessible full texts). The quality of included studies was then assessed using a modified Critical Appraisal Skills Programme (CASP) checklist, adapted to evaluate the methodological soundness and relevance of cybersecurity research. For categorization, thematic analysis was performed iteratively; we initially employed an inductive approach to identify recurring concepts and then used these insights to develop and refine our novel multidimensional taxonomy (Section 4), which subsequently served as a deductive framework for systematic classification and synthesis. This comprehensive structured approach led to the inclusion of more than 250 studies for critical synthesis, significantly enhancing the transparency and auditability of our findings.

Table 1
Related surveys on cyber threat intelligence.

Survey	Context	Contributions
Wagner et al. (2019)	Cyber threat intelligence sharing	• Discusses sharing techniques of CTI • Explores automating CTI sharing
Tounsi and Rais (2018)	Technical threat intelligence	• Classifies and distinguishes among CTI types • Discusses challenges in CTI sharing
Sun et al. (2023)	Cyber threat intelligence mining	• Comprehensive review of latest research on CTI mining from multiple sources • Covers CTI sharing and capabilities
Abu et al. (2018)	Cyber threat intelligence issues and challenges	• Reviews existing research on CTI • Identifies CTI products and services
Samtani et al. (2020)	Cyber threat intelligence platforms	• Systematic review of existing CTI platforms in industry
Schlette et al. (2021)	Cyber threat intelligence and incident response	• Examines incident response from a broader security perspective • Assesses current standardization approaches
Cascavilla et al. (2021)	Cyber crime threat intelligence	• Provides a dual taxonomy for cyber-crime threat intelligence • Systematically analyzes techniques for cyber crime detection
Alaeifar et al. (2024)	Cyber threat intelligence sharing	• Comprehensive study on CTI sharing • Discusses various aspects of CTI sharing
El-Kosairy et al. (2023)	Cyber threat intelligence using blockchain	• Explores how blockchain can enhance CTI sharing
Naseer (2024)	Machine learning and cyber threat intelligence	• Examines the use of machine learning to improve cybersecurity strategies • Discusses visualization of CTI
Chatziamanetoglou and Rantos (2024)	Machine learning in cyber threat intelligence	• Systematic review of recent trends in ML for CTI
Sauerwein et al. (2017)	Cyber threat intelligence platforms	• Systematic review of CTI sharing platforms • Analyzes state-of-the-art vendor landscape

Table 2
Comparative analysis of related surveys on cyber threat attribution.

Survey	Focus area	Key contributions	Limitations of prior work	Gap addressed by our survey
Nisioti et al. (2018)	Intrusion detection and attribution	• Reviews unsupervised and hybrid IDS techniques • Discusses intrusion correlation and basic attribution	• Emphasizes detection; limited treatment of attribution beyond initial correlation	• Delivers a comprehensive review across the full attribution lifecycle—from tactical signals to legal implications
Sharma et al. (2023)	APT attribution	• Analyzes APT evolution and case studies • Surveys attribution methods for APTs	• Narrow focus on APTs; excludes threats like ransomware, cybercrime, and insider attacks	• Covers a broad spectrum of threat actors and attribution contexts, including non-APT threats
Tang et al. (2022)	Threat profiling for APTs	• Reviews intelligent threat profiling techniques • Addresses data, models, and applications for APTs	• Limited to specific profiling methods; lacks generalizability across threat types	• Proposes a generalizable taxonomy for attribution across actor types and operational scales
Liu et al. (2018)	Insider threat detection	• Surveys detection techniques and insider typologies	• Focuses narrowly on insider threats; overlooks external attribution mechanisms	• Integrates insider threats within a unified attribution framework spanning internal and external actors
Shamsi et al. (2016)	Legal and technical attribution	• Analyzes attribution techniques and legal challenges	• Lacks recent advances; limited focus on AI/ML or interdisciplinary integration	• Critically evaluates state-of-the-art AI/ML methods and synthesizes legal, geopolitical, and technical dimensions

1.4. Structure of the survey

This survey systematically reviews cyber threat attribution, beginning with an in-depth analysis of the evolving threat landscape (Section 2) and its inherent challenges (Section 3) across technical, resource, legal, and geopolitical dimensions. We then introduce a novel multidimensional taxonomy (Section 4) to classify attribution efforts, analyze datasets and features (Section 5), critically evaluate existing frameworks (Section 6), and detail cutting-edge data-driven approaches (Section 7), especially those powered by AI and machine learning. Discussions and implications have been presented in Section 8. The paper concludes with strategic implications and a roadmap for future research (Section 9) for practitioners and policymakers.

2. Cyber threat landscape and evolution

The contemporary cyber threat landscape is a dynamic and ever-evolving ecosystem, characterized by a wide array of threats originating

from diverse actors with varying motivations and capabilities (Ahmad and Iqbal, 2024). As depicted in Fig. 2, this landscape encompasses attacks launched by both state-sponsored and non-state actors, ranging from individually operating hackers seeking financial gain or notoriety to organized criminal groups and nation-states pursuing political or strategic objectives (Li, 2017). Also the process followed when formulating the threat taxonomy is depicted in Fig. 1.

2.1. Strategic threat scaling: From criminal opportunism to nation-state precision

Cyberattacks have escalated significantly in scope and impact, migrating from predominantly financially motivated, opportunistic campaigns to sophisticated state-sponsored operations with profound geopolitical intent. Traditional threats like ransomware have evolved into robust, service-based ecosystems such as Ransomware-as-a-Service (RaaS) (Radhakrishnan et al., 2019), enabling scalable deployment by actors with minimal technical expertise. Our analysis reveals that

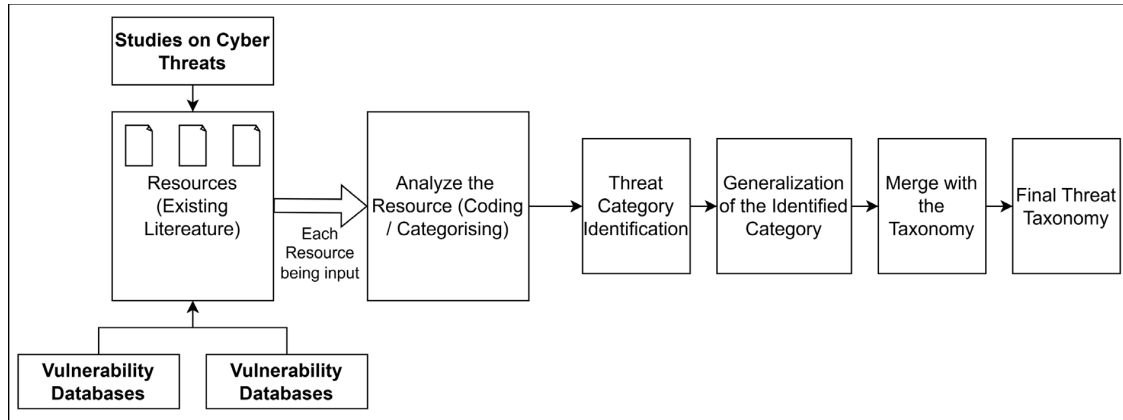


Fig. 1. Threat taxonomy development process.

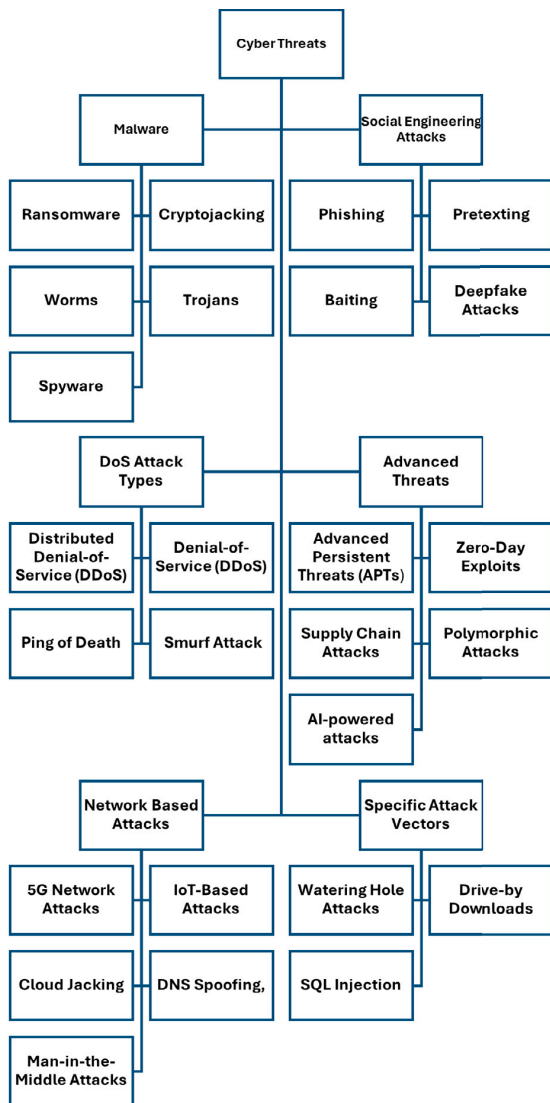


Fig. 2. Threat taxonomy.

ransomware syndicates now incorporate nation-state developed tools, blurring traditional actor boundaries. Simultaneously, Advanced Persistent Threat (APT) groups, historically state-affiliated, are increasingly adopting criminal monetization tactics, exemplified by cryptocurrency heists linked to nation-state actors. Supply chain attacks, such as the SolarWinds incident, further illustrate a significant leap in attacker ambition, compromising trusted software infrastructure to penetrate numerous high-value downstream targets (Heinbockel et al., 2017).

The strategic scaling of cyber threats raises serious attributional challenges. The pervasive use of intermediaries, proxy infrastructure, and sophisticated deception tactics including false flags and intricate misattribution layers allows actors, particularly nation-states, to operate within plausible deniability. This dynamic makes attribution not only a technical process but a complex geopolitical calculus, where the potential for diplomatic repercussions and escalations significantly influences investigative and public disclosure decisions (Krombholz et al., 2015; Ponnumani and Partheeban, 2022). The decentralized nature of cryptocurrencies, often used in ransomware, coupled with mixers and tumblers, further complicates financial tracing (Willard, 2015). The lack of international consensus on norms for attribution and response continues to undermine accountability and deterrence.

2.2. Technological sophistication: AI, automation, and adaptive malice

The increasing incorporation of artificial intelligence (AI) and machine learning (ML) into offensive operations has fundamentally accelerated the complexity of threats. Attackers are increasingly deploying malware that adapts autonomously, intelligent botnets, and algorithmically optimized phishing campaigns (Fritsch et al., 2022; Kaloudi and Li, 2020). Our analysis reveals that fake biometrics, including voice and video, have blurred the lines between reality and fabrication, and weaponizing disinformation for psychological and political impact (Gupta et al., 2023; Takale et al., 2024; Teichmann, 2023). Furthermore, malware variants generated with large language models (LLM) exhibit telltale linguistic patterns within their code that are intentionally designed to mislead attribution engines, creating entirely new obfuscation vectors.

These AI-enhanced threats are significantly more difficult to detect and even harder to attribute. Attack vectors can mutate in real time, attacker intent is deeply obfuscated, and human agency may be indirectly embedded in machine behavior. Attribution must now contend with algorithmic decision-making, opaque training data, and adversarial learning principles. This requires a radical shift toward attribution models that integrate advanced behavioral analytics, forensic linguistics, and machine-in-the-loop traceability, moving beyond traditional signature-based methods (Safitri et al., 2023; Varbanov, 2024).

2.3. Structural vulnerabilities: IoT, ubiquity, and attack surface expansion

The exponential growth of connected devices, especially Internet of Things (IoT) ecosystems, has significantly expanded the attack surface across critical infrastructure, consumer environments, and national defense systems (Shah and Sengupta, 2020). These devices, often deployed with minimal security oversight and weak protocols, frequently serve as both initial attack vectors and force multipliers for large-scale campaigns such as distributed denial of service (DDoS) attacks or botnet propagation (Ali et al., 2020). The ubiquitous nature of IoT devices, coupled with the blurring of lines between cyber tactics and traditional military operations in Hybrid Warfare (Rugge, 2018), further complicates the attribution landscape.

In these cases, the distribution is deeply impeded by device heterogeneity, decentralized control, and poor logging practices within IoT networks. IoT systems often lack standardized telemetry, preventing comprehensive forensic analysis. In addition, fragmentation of ownership and responsibility between manufacturers, users, and integrators creates additional opacity in tracking attacks to their origin. Addressing such challenges requires cross-sector collaboration, enhanced visibility protocols, and regulatory mechanisms to enforce baseline security standards across the vast, distributed IoT landscape.

Implications: Together, these three dimensions strategic escalation, technological sophistication, and structural expansion underscore a critical inflection point in cybersecurity. Attribution is no longer a matter of simply tracing code to origin; it is a multi-domain problem that spans technical, legal, political, and ethical boundaries. As threats evolve into persistent and systemic risks, the attribution challenge becomes not just a question of who but also why, how, and under what strategic logic (Hausken et al., 2024).

To respond effectively, future attribution frameworks must be:

- *Hybrid:* Blending technical telemetry with geopolitical context and behavioral signals.
- *Scalable:* Capable of tracing attacks across distributed systems, including IoT and cloud infrastructures.
- *AI-enabled:* Employing machine learning for real-time threat clustering, pattern detection, and actor inference.
- *Normatively grounded:* Supported by international frameworks for cooperation, evidence-sharing, and escalation management.

3. Challenges of threat attribution

The cyber threat attribution landscape is riddled with multifaceted challenges that span technical, resource-based, legal, geopolitical, and emerging threat-related dimensions. As shown in Fig. 3, these challenges are interdependent and overlap in several dimensions. Understanding these challenges is crucial for developing strategies to overcome them, enhance attribution capabilities, and chart the future directions of attribution (Tsagourias and Farrell, 2020). These challenges and their indicative solutions have been categorized on the basis of extensive literature analysis and synthesis using the process depicted in Fig. 4.

3.1. Technical challenges

The technical landscape of cyber threat attribution is a continuously evolving battleground, where adversaries deploy increasingly sophisticated methods to obscure their identities, evade detection, and subvert forensic analysis. This domain is marked by a dynamic interplay of innovation and countermeasure a cat-and-mouse paradigm driven by three interlocking layers of complexity: *obfuscation and deception*, *fragmented evidence chains*, and *the limitations of current analytical infrastructures and models*. Table 3 shows technical cyber threat attribution challenges and their traditional solutions in the literature.

Obfuscation and Deception as Strategic Shields: Obfuscation and deception techniques have become essential tools for adversaries seeking

to manipulate attribution outcomes. The widespread deployment of anonymization technologies such as VPNs, proxy servers, the Tor network (Čerget and Hudec, 2023), and encrypted peer-to-peer systems has operationalized anonymity as a default protective mechanism. These techniques are no longer just technical workarounds; they represent strategic interventions, exploiting the inherent ambiguity of cyber artifacts to mislead investigators. False flag operations further complicate this landscape by deliberately injecting deceptive indicators into attack payloads, thereby creating a signal inversion problem—where forensic markers are intentionally falsified to confuse attribution efforts (Griffin and Cobb, 2023; Finlay and Payne, 2019).

Attribution in such contexts must thus transcend traditional packet-level forensics and IP tracing. Instead, it demands intent-aware contextual analysis that draws upon socio-technical signals such as linguistic artifacts, temporal attack correlations, infrastructure reuse, and behavioral profiling across campaigns. Only through such multi-modal synthesis can analysts begin to disambiguate strategic deception from authentic attacker signatures.

Fragmented Attribution Chains and Jurisdictional Entanglement: Modern cyber operations frequently unfold across multi-jurisdictional and decentralized infrastructures, including botnets, cloud services, compromised servers, and proxy chains (Kesari et al., 2017; Dalins et al., 2018). These distributed architectures result in fragmented evidence chains, where no singular vantage point affords a comprehensive or coherent attribution path. Moreover, adversaries often exploit legal and diplomatic asymmetries – strategically launching attacks through jurisdictions with weak or non-cooperative legal frameworks – to evade prosecution and frustrate international collaboration (binti Mohamed, 2013; Curtis and Oxburgh, 2022).

Even with robust telemetry and intrusion detection systems, forensic visibility remains bounded by political and institutional constraints. These limitations underscore the urgent need for interoperable institutional mechanisms, such as real-time cross-border data sharing, harmonized evidentiary standards, and collaborative intelligence fusion frameworks that transcend national silos.

Analytical Limitations: Scale, Fragility, and Anti-Forensics: Despite advancements in AI-enabled and behavior-based attribution methods (Prasse et al., 2017), current analytical frameworks are constrained by issues of data scale, model fragility, and anti-forensic tactics. The overwhelming volume of telemetry data generated during attacks often outpaces available analytic capacity (Munappy et al., 2019), while static or heuristics-based machine learning systems struggle to generalize across novel or evolving TTPs (Soliman, 2022). Additionally, these systems are frequently trained on biased or incomplete datasets, limiting their reliability in high-stakes attribution scenarios (Tsagourias and Farrell, 2020; Mehrabi et al., 2022).

Adversaries exacerbate these issues through advanced anti-forensic techniques—including polymorphic or metamorphic malware, encryption, log manipulation, and volatile memory attacks that are specifically designed to erase, corrupt, or misrepresent digital traces (Brezinski and Ferens, 2023; Selamat et al., 2016; Tolley et al., 2021). These tactics erode the evidentiary basis of attribution and amplify the risk of false positives or strategic misattribution. To overcome these challenges, the field must transition from static detection models toward probabilistic and adversarially robust frameworks—capable of incorporating uncertainty, evaluating multiple competing hypotheses, and integrating soft signals such as geopolitical context and actor intent. Such models not only enhance technical attribution but reposition it as a strategic intelligence function—a necessary evolution in the face of sophisticated, multi-layered cyber threats.

Key Trends and Promising Approaches: The technological trajectory of cyber threat attribution is shifting from deterministic, rule-based systems toward more adaptive, intelligence-driven architectures that integrate AI, threat intelligence fusion, and probabilistic reasoning. This fundamental change is a direct response to the escalating complexities

and multifaceted challenges detailed throughout this survey. One dominant trend is the rise of machine learning-powered behavioral analytics, which focuses on identifying adversary TTPs rather than relying on static indicators like IP addresses or malware hashes (Sun et al., 2023; Irshad and Siddiqui, 2024; Rawat et al., 2021). This shift is promising because it enables detection of novel threats, polymorphic attacks, and adversaries employing deception (Nakhodchi et al., 2021; Bilen and Özer, 2021).

Another emerging approach is the use of graph-based and knowledge graph models that correlate disparate data – ranging from malware signatures to geopolitical events – to uncover hidden relationships and adversarial campaign linkages (Lindsay, 2015; Iorga et al., 2020). These models, when coupled with probabilistic inference engines, allow for reasoning under uncertainty, a critical capability in environments with partial or misleading evidence (Bendiek and Schulze, 2021). Federated learning is also gaining attention as a privacy-preserving technique for attribution across multiple organizations or jurisdictions without direct data sharing, directly addressing legal and ethical constraints previously discussed (Ammara et al., 2024; Sikos, 2020). Meanwhile, the integration of cyber threat intelligence (CTI) with natural language processing (NLP) tools allows analysts to extract and codify attribution-relevant signals from unstructured sources like threat reports, underground forums, and social media (Noor et al., 2019; Wang et al., 2021; Ren et al., 2022).

Despite these advances, the adoption of these technologies must be coupled with epistemological scrutiny—ensuring interpretability, auditability, and resistance to adversarial inputs (Skopik and Pahi, 2020; Mehrabi et al., 2022). The overarching implication is clear: effective attribution will increasingly depend on hybrid human–AI systems that balance analytical power with contextual judgment, enhancing both technical accuracy and strategic decision-making (Colbaugh and Glass, 2011; Alghamdi, 2021; Varbanov, 2024). This synthesis underscores the ongoing imperative for continuous innovation and interdisciplinary collaboration to navigate the evolving cyber threat landscape.

3.2. Resource and expertise challenges

The persistent scarcity of resources and expertise remains a fundamental obstacle in achieving accurate and timely cyber threat attribution, particularly for smaller organizations with constrained budgets (Cobb, 2016; Goupil et al., 2022). This challenge is notably dynamic: a pronounced cybersecurity skills gap continuously undermines the capacity to perform attribution tasks requiring specialized knowledge in network forensics, malware reverse engineering, and threat intelligence analysis (Blažič, 2021). This gap is not static; it widens as cyber threats evolve rapidly, outpacing the ability of existing educational and training frameworks to adequately prepare professionals. As attackers increasingly target under-resourced Small and Medium-sized Enterprises (SMEs) as entry points into larger networks, this uneven distribution of human capital exacerbates systemic vulnerabilities and complicates collective defense (Chidukwani et al., 2022).

Even when skilled personnel are available, many organizations lack access to the advanced technological infrastructure necessary for effective attribution, such as malware analysis sandboxes, machine learning-driven analytics, and scalable network forensics platforms (Chidukwani et al., 2022; Saha and Anwar, 2024). This technological exclusion reinforces a critical disparity between large, well-funded institutions and resource-constrained entities (Holland and Burchell, 2022). High-profile breaches, including the Yahoo and Sony Pictures attacks, NotPetya, and Operation Cloud Hopper, powerfully illustrate how limited technical capability and human capital compound attribution has failed (Daswani and Elbayadi, 2021; Ponce et al., 2020; Fayi, 2018). These incidents reveal that under-resourced entities are not only frequent victims but also often serve as crucial entry points into broader digital ecosystems, significantly aggravating systemic vulnerabilities and hindering accurate attribution for complex, multi-stage campaigns (Chidukwani et al., 2022).

Implications: The implications of these resource and expertise limitations are profound: inadequate expertise and limited access to advanced tools directly hinder timely and accurate attribution, reduce the effectiveness of deterrence, and amplify the overall impact of sophisticated attacks. Addressing this challenge demands a coordinated, multi-pronged response. Promising approaches emphasize multi-stakeholder collaboration, including significantly scaling up investment in continuous cybersecurity education and professional training (Spidaleri and McArdle, 2016); encouraging robust information-sharing ecosystems that bridge public–private divides and empower smaller players (Cook, 2017); and proactively lowering barriers to attribution tools through affordable, cloud-based, or open-source alternatives (Cook, 2017). Moreover, instituting increased government and industry support through funding, technical assistance, and policy frameworks is crucial to offset structural imbalances and foster an equitable, resilient attribution ecosystem (Lee, 2014). Without such reforms, the attribution landscape risks remaining skewed and reactive, severely limiting its deterrent and forensic value, as summarized in Table 4.

3.3. Legal and geopolitical challenges

The attribution of cyber threats operates within a complex interplay of legal, technical, and geopolitical factors, presenting persistent challenges for investigators and policymakers (Tsagourias and Farrell, 2020; Yannakogeorgos, 2013). These barriers stem from the transnational nature of cyberspace, where attacks often traverse multiple jurisdictions, rendering traditional legal frameworks and investigative tools inadequate (binti Mohamed, 2013; Razmetaeva et al., 2021). Table 5 presents traditional approaches of legal and geopolitical cyber threat attributions.

Legal and Jurisdictional Complexities in Cyber Attribution: Cross-border cyber investigations routinely encounter jurisdictional conflicts, as states uphold differing legal systems, data protection regimes, and evidentiary requirements (Iftikhar, 2024; Brenner, 2004). Mechanisms like Mutual Legal Assistance Treaties (MLATs), although foundational for international cooperation, are frequently slow, procedurally burdensome, and hampered by political constraints (Shipley and Bowker, 2014; Curtis and Oxburgh, 2022). In the 2020 SolarWinds breach, for example, the attribution process was slowed by limitations in international cooperation and data access across borders. Moreover, the lack of a globally accepted cybercrime jurisdiction framework leaves legal grey zones that can be exploited by malicious actors operating across safe havens (Varol and Sonmez, 2021; Assembly, 2022). These issues are exacerbated by inconsistent evidence handling standards, which compromise the admissibility and integrity of forensic data.

To address these deficiencies, legal harmonization through multi-lateral agreements—such as expanding participation in the Budapest Convention—remains vital (Didenko, 2020; Iu and Wong, 2024). However, global adoption is limited by political divergence, particularly among major cyber powers. Enhancing cooperation via faster, digitalized MLAT platforms and fostering trust through regional cyber diplomacy, joint attribution exercises, and standardized evidence formats could improve investigative efficiency (Chen et al., 2018; Wagner et al., 2019; Dykstra et al., 2023; Ron et al., 2020). Importantly, such frameworks must respect national sovereignty while promoting interoperability and mutual legal recognition.

Geopolitical Influences on Attribution: Geopolitical dynamics heavily influence whether, how, and when cyberattacks are attributed. States often weigh the strategic risks of public attribution – including potential diplomatic fallout, sanctions, or retaliatory cyber operations – against the benefits of transparency (Schulzke, 2018). For instance, in the NotPetya attack attributed to Russian military intelligence, some states hesitated to make official statements due to diplomatic sensitivities. Conversely, attribution can be politicized and used as a tool of statecraft to justify retaliatory measures or shape international opinion (Banks, 2021; Egloff and Wenger, 2019).

These dynamics are compounded by mistrust among states with divergent cyber doctrines—some favoring offensive capabilities over cooperative defense. Such fragmentation undermines joint attribution efforts and weakens deterrence. Confidence-building measures, including bilateral security dialogues, non-binding notification norms, and cyber diplomacy forums, can mitigate tensions and encourage responsible state behavior (Efrony, 2024; Wanis-St John, 2004). While proposals for neutral attribution mechanisms – such as UN-facilitated investigation teams or regional cyber incident response coalitions – have been raised, progress is hindered by geopolitical rivalry and concerns over impartiality (Egloff and Smeets, 2023; Efrony, 2024; Shany and Schmitt, 2020; Shopina et al., 2020). Incremental trust-building through transparency initiatives and shared threat intelligence protocols may provide a more feasible path forward.

Data Privacy and Investigative Trade-Offs: Attribution efforts often conflict with data privacy obligations. Regulations such as the EU's General Data Protection Regulation (GDPR) restrict the collection and cross-border transfer of personal data, complicating evidence gathering and forensic collaboration (Raul, 2019). This creates a structural tension between protecting individual rights and enabling timely investigation. For example, during investigations into the 2018 Marriott breach, investigators faced challenges in aligning data-sharing practices with privacy regulations across jurisdictions.

Emerging privacy-preserving technologies – such as differential privacy and homomorphic encryption – offer partial solutions by allowing certain types of analysis without exposing raw data (Ogburn et al., 2013; Arzberger et al., 2004). However, these techniques can reduce evidentiary granularity and introduce uncertainty into already ambiguous attribution chains. Legal accountability requires not only technical sophistication but also clear evidentiary thresholds that ensure admissibility across diverse legal contexts (Didenko, 2020; Iu and Wong, 2024). To balance these competing interests, attribution frameworks must integrate compliance-by-design approaches that align with regional data laws and establish transparent data handling policies. Policymakers may also consider narrowly tailored exceptions for cyber investigations, under judicial oversight, to enable lawful evidence sharing without compromising fundamental rights.

Trends, Implications, and Promising Approaches: Recent trends in cyber threat attribution reveal a growing convergence of legal, political, and technological dimensions, each shaping how states and institutions respond to increasingly sophisticated attacks. A noticeable shift toward multilateral dialogue, regional cooperation, and the standardization of digital forensics protocols reflects an emerging recognition that attribution cannot be achieved in isolation. While long-term proposals – such as the establishment of an international cybercourt or a UN-sponsored attribution body – remain aspirational, more immediate progress lies in strengthening existing instruments like the Budapest Convention, promoting interoperable digital evidence standards, and fostering regional attribution alliances that share both responsibility and capability.

Technological developments are reshaping attribution practices. AI-driven attribution tools, especially those that offer explainable, auditable decision-making processes, hold promise for accelerating investigations and reducing bias. However, such tools must be used cautiously. Overreliance on automated systems can obscure accountability and amplify the risks of misattribution, particularly in politically sensitive contexts. The 2021 Colonial Pipeline attack underscored how hasty or incorrect attribution could have destabilizing diplomatic consequences.

The global disparity in cyber investigative capacities also has critical implications. Without deliberate investment in training, forensic infrastructure, and secure data-sharing mechanisms, under-resourced countries may remain on the margins of collaborative attribution efforts. This inequity weakens global response coordination and may foster safe havens for malicious actors. Addressing this requires international capacity-building initiatives that are inclusive, sustainable, and technically robust.

Looking forward, promising approaches involve layered, resilient governance: integrating multilateral legal frameworks with bilateral trust-building, establishing neutral and secure technical infrastructure, and embedding ethical oversight across all attribution processes. The credibility and deterrence power of cyber attribution will ultimately depend not only on technical accuracy but also on procedural transparency, political neutrality, and global inclusivity.

4. A multidimensional taxonomy of cyber threat attribution

Cyber threat attribution is not merely a technical exercise but a complex, multidisciplinary endeavor to identify and hold accountable perpetrators, understanding the “who”, “what”, “how”, and “why” of cyberattacks (Warikoo, 2021; Noor et al., 2019; Lindsay, 2015; Lin, 2016b). To provide actionable intelligence for defensive strategies and legal actions (Irshad and Siddiqui, 2023), this process demands a precise, multidimensional classification. The Interdisciplinary Attribution Matrix (IAM) is proposed as a novel conceptual framework that systematically organizes attribution research and incidents across distinct, orthogonal axes, directly addressing prior limitations concerning overlapping categories. This synthesized taxonomy, conceptually depicted in Fig. 6, allows for a structured understanding of different attribution methodologies and their intricate relationships, with its development process akin to that depicted in Fig. 5.

4.1. Attribution confidence & granularity (the “level” of attribution)

This axis reflects the increasing degrees of certainty and the specificity of the attributed entity, determining the potential for actionability. These hierarchical levels range from immediate technical identification to legally actionable insights, each serving distinct objectives and requiring varying evidence types and confidence thresholds, as summarized in Table 6.

Tactical Attribution primarily involves identifying immediate technical indicators of compromise (IOCs), such as malware signatures, IP addresses, and attack patterns (Wang et al., 2021; Vehabovic et al., 2023; Kim et al., 2021). This serves as the initial entry point, providing crucial information for rapid incident response, identification of compromised systems, and development of immediate mitigation measures. At this level, analysts employ Malware Analysis, dissecting malicious code through static and dynamic techniques to identify unique patterns, programming styles, and infrastructure links (Singh and Singh, 2021; Lee et al., 2023; Megira et al., 2018). Concurrently, Network Traffic Analysis scrutinizes digital trails for anomalies, communication channels, and data exfiltration routes to reconstruct attacker actions (Shandilya et al., 2023). Digital Forensics meticulously collects, preserves, and analyzes digital evidence from compromised systems to reconstruct attacker actions (Kara, 2021; Bhandari and Jusas, 2020). While essential for immediate response, tactical attribution offers a narrow view of the threat landscape, focusing on the “how” without revealing underlying motivations or broader strategic objectives, akin to analyzing a weapon without understanding the perpetrator's intent.

Operational Attribution builds upon the tactical level by seeking to identify the specific group or organization responsible for an attack (Wang et al., 2021; Vehabovic et al., 2023; Kim et al., 2021). This involves analyzing the consistent Tactics, Techniques, and Procedures (TTPs) employed by adversaries, examining their broader infrastructure, and assessing their potential motivations. By linking these elements to known threat actors or groups, operational attribution provides valuable insights into their capabilities, intent, and potential future actions. This level relies heavily on contextual intelligence, drawing from Threat Intelligence Feeds to correlate observed attack patterns with known actor TTPs (Meier et al., 2018; Griffioen et al., 2020; Pastor-Galindo et al., 2024), Social Media Monitoring for insights into attacker motivations and affiliations (Huang and Ban, 2020; Alevizopoulou et al., 2021), Dark Web Forums for discussions on

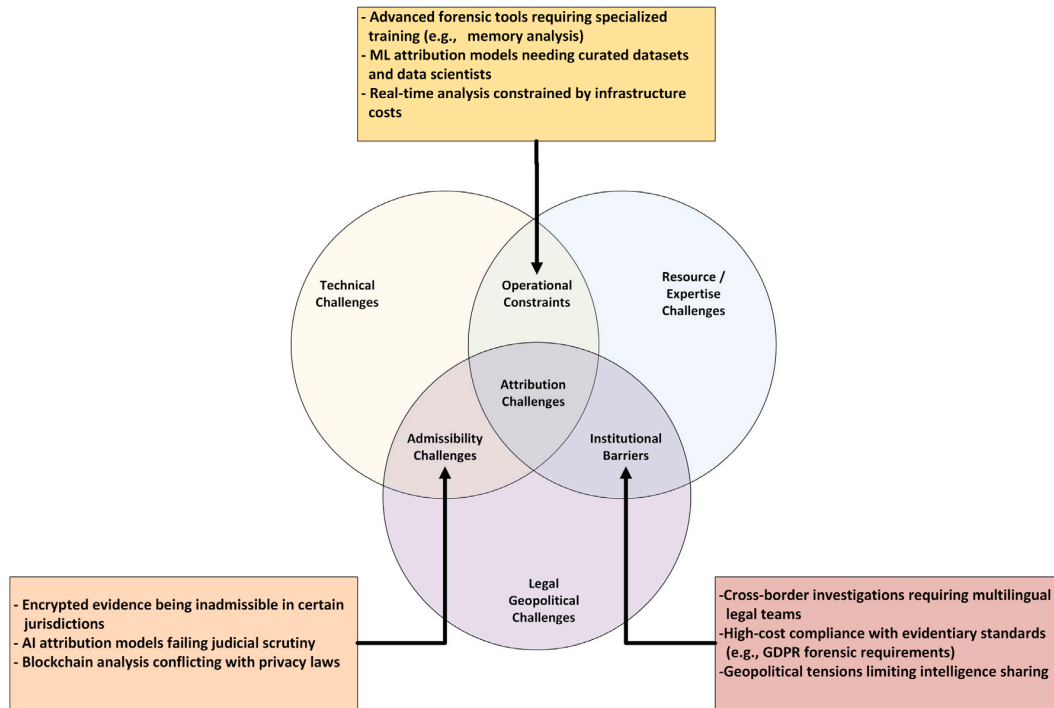


Fig. 3. Interdependencies and overlap of cyber attribution challenges.

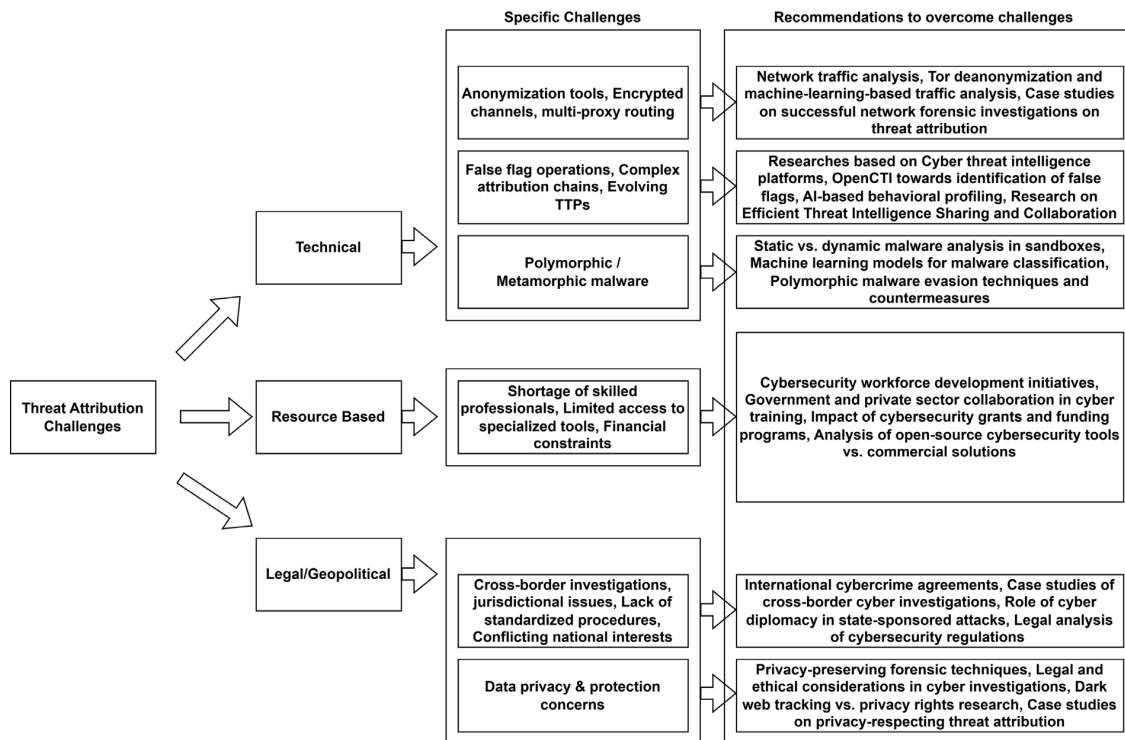


Fig. 4. Solution identification process.

vulnerabilities and criminal activities (Schäfer et al., 2019; Ngo et al., 2023), and Technical Forums and Blogs for emerging threat information (Holtz et al., 2012). While providing a deeper understanding,

operational attribution largely focuses on the “how” and “who” at the group level, and can be limited by information on newly emerging groups or those employing sophisticated evasion.

Table 3
Technical challenges in cyber threat attribution.

Specific challenge	Impact on attribution	Solutions in the literature	References
Anonymization tools (VPNs, Tor, proxy servers)	Obfuscation of attacker origin	• Advanced network forensics and machine learning: Detects patterns in anonymized traffic through techniques such as traffic correlation, onion routing analysis, and Tor deanonymization. These approaches can be used to exploit timing, traffic fingerprinting, time-correlation attacks, packet size, and endpoint correlations to infer origin points even within anonymized networks which could reveal patterns of movement across obfuscated nodes. • Machine learning algorithms: ML algorithms aid in classifying encrypted or anonymized traffic and correlating behaviors across sessions. This allows for probabilistic inference of origin, even without decrypting payloads. Also ML can be used for anomaly detection.	Javed et al. (2024), Simioni et al. (2021), Ji et al. (2017), Hansen (2013), Nasr et al. (2018) and Karunanayake et al. (2021)
False flag operations	Misdirection of investigation	• Behavioral analysis: Makes it possible to profile consistent attacker TTPs, which are difficult to fake over time even if the attacker uses false flags. • Threat intelligence platforms: Collect data across sources to correlate activities and expose inconsistencies. The multi-source data from different platforms neutralizes the application of false flags. • Multi-factor attribution: Combines different factual data such as behavioral, contextual, and cross-dataset analysis to reveal planted evidence. Combining multiple independent attributes helps uncover inconsistencies and leads to correct attribution since the attackers find it hard to apply false flags in multiple sources.	Skopik and Pahi (2020), Pahi and Skopik (2019), Bartholomew and Guerrero-Saade (2016) and Hwang and Kim (2023)
Complex attribution chains	Difficulty in tracing source	• Network mapping: Sophisticated attackers often use layered infrastructures such as botnets, chained proxies etc. Network Mapping reveals infrastructure paths across intermediary systems which paves the way for tracing and correlation of the infrastructure to known historic attacks. • Malware analysis: Analysis of malware samples tied to stages in the chain can expose pivoting behaviors. • Collaboration: Collaboration with other entities such as organizations, nation states, security providers etc. improves visibility and enables source triangulation to track the entire chain.	Shany and Schmitt (2020), Riccardi (2016), Leevy et al. (2021), Li (2024), Garetto et al. (2003), Chen and Ji (2005) and Qiu et al. (2024)
Evolving TTPs (Tactics, Techniques, and Procedures)	Inability to keep up with threats	• Continuous monitoring and threat intelligence sharing: Continuous monitoring using endpoint detection tools and threat intelligence sharing allows near real-time updates of known TTPs which can be utilized for attribution of new threats. • Proactive defense: The use of proactive defense strategies adapts quickly to emerging trends, allowing rapid integration into defensive and attribution systems.	Zeming et al. (2015), Aminu et al. (2024) and Torkura et al. (2021)
Encrypted channels, multi-proxy routing	Enhanced attacker stealth	• Traffic analysis and decryption tools: Traffic analysis and legally viable decryption tools help to expose communication metadata. • International cooperation: Enables access to logs across jurisdictions to incorporate global level indicators to the attribution process to speed up gaps in required information. • Deanonymization: Deanonymizing patterns across proxies which is feasible via timing attacks and machine learning on encrypted metadata. Deanonymizing techniques such as traffic timing allow observing similar traffic timing at the entry and exit nodes which enables correlation of the user to the destination.	Liberatore and Levine (2006), Arikkat et al. (2024), Papadogiannaki and Ioannidis (2021), Alserhani (2024), Maartmann-Moe et al. (2009) and Pannu et al. (2016)
Polymorphic or metamorphic malware	Evasion of detection	• Behavior based detection: Can be utilized focusing on runtime behavior rather than static signatures which bypass code obfuscation. • Sandboxing and ML-based execution profiling: Helps detect even transformed variants by analyzing API calls, memory usage, and process behavior.	Kuriyal et al. (2022), Popli and Girdhar (2019), Qu and Hughes (2013), Ali and Ali (2020) and Badhwar (2021)

Strategic Attribution transcends technical details, aiming to understand the broader context, motivations, and potential long-term impact of cyber operations (Warikoo, 2021; Noor et al., 2019; Puzis and Angappan, 2020; Avellaneda et al., 2019). This level seeks to answer the “why” question, uncovering underlying reasons and strategic objectives to inform national cybersecurity strategies, policy decisions, and international cooperation. Analysis involves assessing Motivations and Goals by examining geopolitical factors, economic interests, and ideological beliefs driving a cyber operation, and performing comprehensive Impact Assessment on national security and critical infrastructure. The strategic goal is Long-term Threat Analysis, anticipating future attack vectors and developing proactive defense measures. The scarcity of explicit research focused solely on strategic attribution highlights a critical gap in the current understanding of cyber threats.

Legal Attribution represents the highest and most challenging level, requiring evidence that meets stringent legal standards for prosecution or formal diplomatic action (Warikoo, 2021; Skopik and Pahi, 2020). This level is crucial for effective deterrence and upholding the rule of law. It demands meticulous Evidence Collection and Analysis, adhering to strict Chain of Custody protocols to ensure integrity and admissibility. Close Collaboration with Law Enforcement agencies and International Cooperation through mechanisms like Mutual Legal Assistance Treaties (MLATs) are indispensable for cross-border evidence gathering and bringing suspects to justice. The process is lengthy and complex due to international law, diverse evidentiary standards, and the elusive nature of cyber attackers, necessitating a multidisciplinary approach bridging technical expertise with legal and political science perspectives.

Table 4
Resource based challenges in cyber threat attribution.

Specific challenge	Impact on attribution	Proposed solutions	References
Shortage of skilled professionals	Limited investigation capacity	• Education and Training: The sufficient education and training initiatives develop a pipeline of skilled cybersecurity analysts. Investing in cybersecurity education, certification programs, and academic-industry partnerships builds long-term capabilities. • Collaboration: The collaboration and resource sharing maximize existing expertise across organizations, especially for smaller or under resourced entities. Collaborative networks like ISACs and joint task forces help less-resourced organizations leverage external expertise on-demand. • Partnerships: Public private partnerships and mentorship programs can accelerate capacity building.	Cobb (2016), Blažič (2021), Crumpler and Lewis (2022), Angafor et al. (2020) and Nguyen (2024)
Limited access to specialized tools and technologies	Inability to analyze complex data	• Utilizing free tools and Resources: Certain open source tools, community driven tools and cloud-based platforms democratize access to forensic, ML, and threat analysis technologies while reducing the dependency on expensive proprietary systems. • External support: Support from government and industry through funding or technical assistance fills gaps in under resourced organizations and ensures wide availability of core capabilities. The National security programs can provide subsidized access to premium capabilities.	Andreatos (2023), Chidukwani et al. (2022) and Tam et al. (2021)
Financial constraints in smaller organizations	Vulnerability to attacks	• Financial aid and tax incentives: The financial aid received via grants, subsidies, or tax incentives enables smaller organizations to invest in cybersecurity infrastructure, training, and threat intelligence tools while lowering the barrier for adoption of best practices and technical strength. It reduces their exposure and improves attribution capabilities	Gordon and Smith (2007), Kosseff (2016) and Alderson and Hoo (2004)

Table 5
Legal & Geopolitical challenges in cyber threat attribution.

Specific challenge	Impact on attribution	Proposed solutions	References
Cross-border investigations & jurisdictional issues	Delays, disputes, inaction	• Legal frameworks and MLATs: The Standardized legal frameworks and Mutual Legal Assistance Treaties (MLATs) facilitate evidence sharing across borders and improve international investigative cooperation. Also it harmonizes procedures and mitigates legal bottlenecks. • Cross border investigation frameworks and Data sharing agreements: Cross border investigative frameworks, and data sharing agreements allows to facilitate faster cooperation and legally sound evidence exchange.	Koops and Goodwin (2014), Bukola (2019) and Swire and Hemmings (2015)
Lack of standardized procedures	Inconsistent investigations	• Establishment of international norms and protocols: Without common frameworks, attribution varies by organization and country. The establishment of international norms and protocols enables consistency in how attribution is approached across nations and sectors which helps to resolve jurisdictional information retrieval conflicts. • Information sharing and coordinated practices: Information sharing and coordinated practices improve reliability and reproducibility. Standard operating procedures, frameworks like NIST or ISO, and alignment with platforms such as the Budapest Convention improve comparability, repeatability, and global trust in attribution outcomes	Koops and Goodwin (2014), Bukola (2019) and Swire and Hemmings (2015)
Conflicting national interests & sensitivities	Reluctance to attribute	• Confidence, diplomacy and transparency building measures: The Geopolitical rivalries can make states reluctant to formally attribute attacks. Confidence building measures such as cyber diplomacy, bilateral security dialogues, non-binding incident notification norms, transparency agreements, and backchannel communication foster trust and allow collaborative action even in politically tense contexts.	Lord (2012), Lindley (2021) and Finnemore and Hollis (2016)
Data privacy & protection concerns	Limited access to data	• Regulations enforcement: The regulations like GDPR restrict sharing personal or sensitive data. • Privacy preserving analytics: Privacy preserving analytics such as homomorphic encryption, differential privacy etc. allow data analysis without exposing sensitive information. These techniques ensure compliance with regulations during the attribution process.	ReddyAyyadapu (2022), Keshk et al. (2019) and Gopireddy (2023)

Table 6
Comparison of attribution based on levels.

	Tactical	Operational	Strategic	Legal
Duration	Short term	Short term	Long term	Long term
Impact	Immediate response and mitigation	Immediate response and mitigation	Long term policy decisions, proactive defense strategies, anticipate future attacks	Accountability and non repudiation, upholding law in cyber space
Objective	Identification of tools and techniques	Identify the specific adversaries	Identify the motivation and goals	Identify legally acceptable evidence and proving perpetrator accountability
View	Narrow	Narrow	Broad	Broad
Elements of analysis	Malware signatures, Network traffic analysis, Attack patterns, digital forensics	Threat intelligence feeds, social media monitoring, web forums	Geopolitical factors, economic factors, ideological factors	Evidence, law enforcement collaboration, international cooperation

4.2. Analytical domains (the “how” and “where” of evidence processing)

This axis categorizes the core analytical methodologies and the primary evidence streams utilized, differentiating the type of analysis performed to extract attribution-relevant intelligence. It defines the distinct lenses through which fragmented digital traces are examined to build a comprehensive picture of a cyberattack.

Digital & Network Forensics stands as a bedrock of attribution, meticulously focusing on technical artifacts extracted from compromised systems—including disk analysis, memory analysis, log parsing, and artifact extraction (Mei et al., 2024; Kara, 2021; Bhandari and Jusas, 2020). This domain also rigorously scrutinizes network traffic, analyzing packets, flows, domain names, and communication patterns to identify Indicators of Compromise (IOCs) and establish links to specific threat actors (Čerget and Hudec, 2023; Shandilya et al., 2023). It aims to reconstruct attack paths, identify C2 servers, and detect anomalies, forming the bedrock of technical attribution (Lin, 2016a). Its strength lies in providing detailed and often legally admissible evidence. However, its effectiveness is frequently limited by sophisticated anti-forensic methods, such as data encryption and log tampering, employed by adversaries.

Malware Analysis provides an in-depth understanding of attacker tools and techniques by dissecting malicious code to understand its functionality, identify origins, and potentially link it to specific threat actors or campaigns (Boot, 2019). This approach encompasses both static analysis (examining code without execution for signatures and characteristics) and dynamic analysis (executing malware in controlled environments to observe behavior). Such analysis can reveal hidden functionalities and provide crucial attribution clues. The continuously evolving nature of malware, with attackers employing techniques like polymorphism and metamorphism to evade detection (Brezinski and Ferens, 2023; Selamat et al., 2016; Canfora et al., 2014), and the rise of fileless malware operating in memory (AliAhmad et al., 2023), pose significant challenges. Research into Natural Language Processing (NLP) to analyze malware code and identify linguistic patterns, alongside the integration of malware analysis with other attribution sources, offers promising future directions (Wang et al., 2021).

Behavioral Analysis focuses on identifying the Tactics, Techniques, and Procedures (TTPs) employed by adversaries, recognizing their unique operational patterns. This involves analyzing various aspects of attacker behavior, such as reconnaissance methods, exploitation techniques, command and control (C2) infrastructure interactions, and data exfiltration strategies. While threat actors continuously adapt their TTPs to evade detection and attribution, making consistent pattern establishment difficult, the increasing use of automation and AI by attackers further complicates analysis by obscuring human fingerprints (Irshad and Siddiqui, 2023; Noor et al., 2019). Nevertheless, Machine Learning and AI are being leveraged to enhance behavioral analysis, with models like “attack2vec” extracting intricate details

from attack patterns and improving the identification of malicious activity. Future research aims for more robust and explainable ML models for behavioral analysis, integrating diverse data sources to build comprehensive threat actor profiles.

Open-Source & Contextual Intelligence (OSINT/GEOINT/FININT) synthesizes information from publicly available sources (social networks, news), geopolitical factors, financial flows, and linguistic analysis to infer motive, origin, and identity not directly observable through purely technical compromise (Ruohonen et al., 2024). Within this domain, Linguistic Analysis scrutinizes subtle linguistic traits embedded in digital footprints, such as code comments or attacker communications, to uncover cultural, regional, or national indicators that point toward a threat actor’s identity (Perry et al., 2019). Temporal Analysis assesses attack timing patterns (e.g., frequency of intrusions, C2 server communication times, or time zones of activity) to infer operational habits and geographical origins (Shenderovitz and Nissim, 2024). Infrastructure Correlation maps reused IPs, SSL certificates, and domains to known campaigns, thereby linking disparate attacks by discovering shared infrastructure or payloads (Saha et al., 2025). This domain provides crucial “why” and “who” insights, significantly complementing technical data, though it is susceptible to data overload and variability in quality.

Emerging AI-Driven Analytics (Cross-Cutting Capability) represents a powerful paradigm applied across all aforementioned analytical domains. Machine Learning (ML) algorithms are leveraged to analyze massive datasets from network logs or malware behaviors, identifying complex patterns and anomalies that correlate to specific actors or campaigns (Irshad and Siddiqui, 2023). This enables the detection and attribution of zero-day attacks and offers scalability with large data volumes and evolving attack vectors. Other foundational techniques increasingly enhanced by AI include Signature-Based detection for known patterns and IOCs (Hashimyar et al., 2025) and Heuristics-Based analysis, which uses rule-based logic to detect and attribute threats based on behavior or structure, even for unknown malware samples (Sahoo, 2022). Threat Intelligence Fusion is further bolstered by AI, combining multiple sources of intelligence (technical, contextual, human, OSINT) for a holistic view of an actor’s identity, capability, and intent (Xiang et al., 2024). However, the adoption of these AI-driven approaches necessitates rigorous epistemological scrutiny to ensure interpretability, auditability, and resistance to adversarial inputs, addressing challenges related to data quality, bias, and model complexity (Noor et al., 2019).

4.3. Adversarial motivation & profile (the “why” and “who” classification)

This axis categorizes the driving force and typical characteristics of the threat actor, which profoundly influence their TTPs, operational security, and the specific challenges they pose for attribution. This framework moves beyond merely identifying the “target” to classifying the perpetrator’s inherent nature.

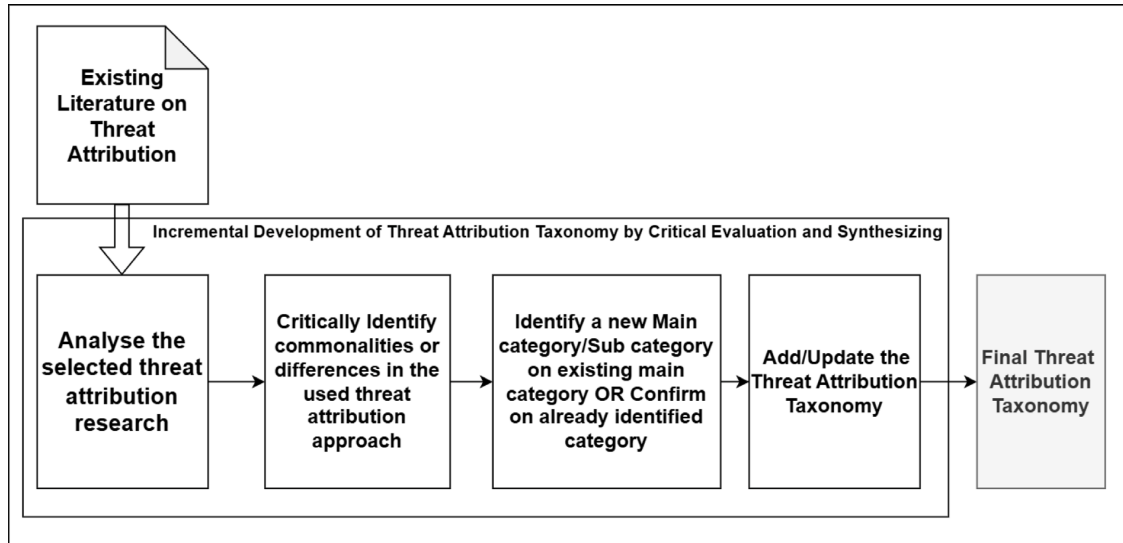


Fig. 5. Threat attribution taxonomy development process.

Financially Motivated (Cybercrime) actors are driven primarily by monetary gain, operating often like businesses with specialized skills and resources (e.g., ransomware syndicates, fraudsters). Attribution is challenging due to their sophisticated operational security, extensive use of anonymization techniques like cryptocurrencies and bulletproof hosting, and encrypted communication channels (Willard, 2015; Bailey et al., 2014). Malware attribution, in this context, focuses on identifying groups distributing malicious software by analyzing code behavior and infrastructure (Wang et al., 2021; Vehabovic et al., 2023; Kim et al., 2021).

State-Sponsored/Geopolitical actors leverage cyber capabilities to achieve national political or strategic objectives, such as espionage, sabotage, or influence operations. These actors possess significant resources, custom tools, and advanced capabilities, frequently employing zero-day exploits and sophisticated false flag operations (Krombholz et al., 2015; Irani et al., 2011). Attribution is complicated by geopolitical considerations and the deliberate intention of plausible deniability, requiring extensive intelligence analysis and a cautious approach due to severe diplomatic consequences (Ponnumani and Partheeban, 2022).

Ideological/Disruptive (Hacktivism) groups are motivated by social, political, or ethical causes, using cyberattacks to promote their agenda or disrupt targets. Their attacks can range from defacements and data leaks to DDoS. Attribution is challenging due to their anonymous operations and reliance on distributed networks (Sailio et al., 2020; Sembera et al., 2021). Tracing hacktivists often relies more on OSINT and social media monitoring than deep technical forensics.

Insider Threats originate from individuals originate from individuals operating within an organization who misuse their authorized access to sensitive information or systems (Mazzarolo and Jurcut, 2020). These threats are particularly dangerous due to privileged access and knowledge of internal vulnerabilities. Attribution is challenging given the subtle and evasive nature of insider actions, which often blend with normal work routines and leave minimal technical traces (Mazzarolo and Jurcut, 2020; Prabhu and Thompson, 2022). Effective attribution for insiders requires a multi-faceted approach, combining internal user activity monitoring, data loss prevention tools, behavioral analytics, and collaboration with human resources (Hunker and Probst, 2011; Al-Mhiqani et al., 2020).

4.4. Taxonomy at work - cases studies

To demonstrate the practical utility and analytical power of the taxonomy, this section applies the framework to dissect two prominent

cyber incidents and two key research studies. This analysis showcases how the IAM provides a more nuanced, structured, and comprehensive understanding of attribution challenges, processes, and implications, offering insights beyond traditional categorization methods. The process of analyzing cases and studies against the IAM involves mapping each incident or research focus onto the specific categories within each of the three axes, as conceptually illustrated in Fig. 7.

4.4.1. Analysis of threat incidents against the IAM

NotPetya Ransomware Attack (2017): The NotPetya attack, initially perceived as a financially motivated ransomware campaign in June 2017, was later definitively identified as a destructive wiper malware operation aimed primarily at Ukrainian institutions, with significant global spillover (Bendiek and Schulze, 2021; Lika et al., 2018; Krasznay, 2020). Applying the IAM reveals its multifaceted nature. Attribution for NotPetya rapidly escalated through multiple levels within Axis 1 (Attribution Confidence & Granularity). Initial analysis achieved Tactical (malware signature, EternalBlue exploit, network traffic patterns) and Operational (linking to Sandworm, a known GRU-affiliated APT group) certainty. The decisive step involved Strategic attribution, identifying the attack's primary motivation as geopolitical pressure, economic sabotage, and a test of cyber warfare capabilities, leading to Legal level attribution through official government statements and diplomatic sanctions against Russia (Egloff, 2020; Krasznay, 2020; Aljohani, 2024). Within Axis 2 (Analytical Domain), attribution relied heavily on Digital & Network Forensics (detailed malware analysis revealing characteristics shared with prior GRU malware, tracing its spread via EternalBlue, and analyzing C2 infrastructure associated with Sandworm). Crucially, Open-Source & Contextual Intelligence (geopolitical tensions, victim targeting patterns, public statements) played a pivotal role in establishing the strategic intent beyond the technical indicators. For Axis 3 (Adversarial Motivation & Profile), NotPetya is a quintessential example of an attack by a State-Sponsored/Geopolitical actor (Russia, GRU). Its true motivation was disruption and political influence, starkly contrasting with typical Financially Motivated (Cybercrime) ransomware, despite its initial disguise. Analyzing NotPetya through the IAM highlights the deliberate use of deception to mask strategic intent, demonstrating how a purely technical view (Axis 2) would fail to capture the political motivation (Axis 3) essential for high-level (Axis 1) attribution.

SolarWinds Supply Chain Attack (2020): The SolarWinds supply chain attack represented one of the most sophisticated cyber espionage

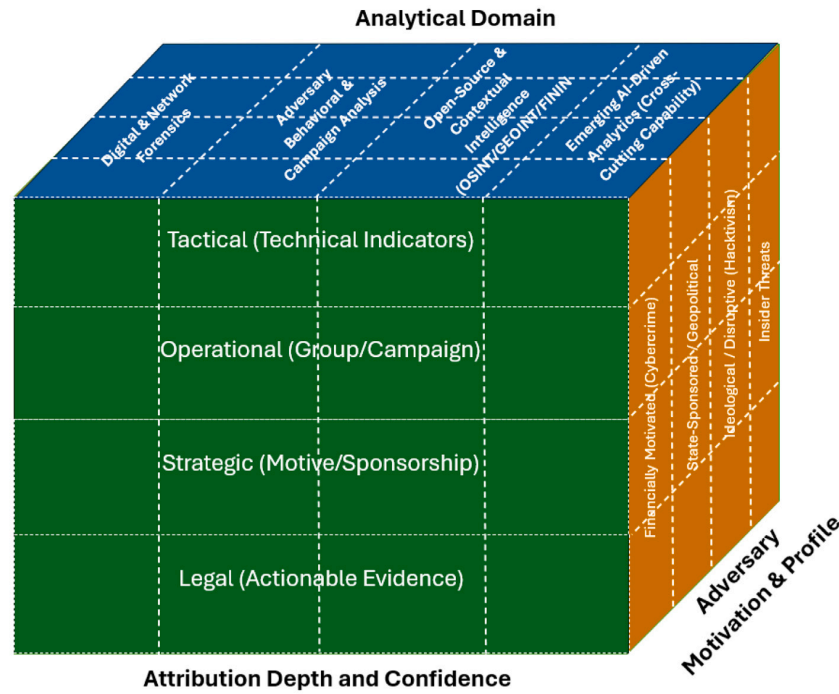


Fig. 6. Threat attribution taxonomy.

operations in history, affecting multiple U.S. government agencies and corporations (Kruti et al., 2023). Its long-term, stealthy nature differentiated it from disruptive attacks like NotPetya. For Axis 1 (Attribution Confidence & Granularity), attribution primarily targeted the Strategic level, identifying the objective as long-term intelligence gathering from high-value targets (U.S. federal agencies, cybersecurity firms). This strategic understanding subsequently underpinned Legal level attribution, leading to U.S. government sanctions against Russian entities. Tactical and Operational analysis (malware analysis, C2 patterns) were foundational, providing the evidence for this higher-level determination. Within Axis 2 (Analytical Domain), extensive Adversary Behavioral & Campaign Analysis was paramount, as the attack's methodology consistently resembled past APT29 (Cozy Bear) cyber intrusions, characterized by stealth and persistence. This was combined with rigorous Digital & Network Forensics (analysis of compromised SolarWinds Orion software updates, injected malicious code, and C2 communications). Open-Source & Contextual Intelligence also played a significant role, correlating findings with geopolitical intelligence assessments. For Axis 3 (Adversarial Motivation & Profile), the attack clearly originated from a State-Sponsored / Geopolitical actor (Russian Foreign Intelligence Service - SVR, linked to APT29/Cozy Bear), with the objective being espionage rather than financial gain or overt disruption. The use of a supply chain compromise underscored the advanced capabilities of this profile. The IAM's application to SolarWinds demonstrates how profiling a highly sophisticated State-Sponsored actor (Axis 3) through Behavioral and Forensic domains (Axis 2) over an extended period is crucial for achieving high-confidence Strategic (Axis 1) attribution that informs national policy.

4.4.2. Analysis of research studies against the IAM

The IAM can also systematically classify and analyze existing research, highlighting their primary contributions and informing future directions.

Study by Egloff (Egloff, 2020): The study by Egloff (Egloff, 2020) primarily explores public attribution as a Strategic tool in cybersecurity policymaking. Its analysis positions attribution at the Strategic level within Axis 1 (Attribution Confidence & Granularity), examining how states leverage high-level attribution to shape political narratives, norms, and international responses to cyber threats. While not primarily focused on specific analytical domains in Axis 2, the study inherently touches upon the profile of State-Sponsored/Geopolitical actors within Axis 3 (Adversarial Motivation & Profile), specifically analyzing the policy implications of attributing their actions. This research highlights the critical shift from merely identifying attackers to the strategic decision-making surrounding public disclosure and its political ramifications.

Study by Leite et al. (2024): The study by Leite et al. (2024) proposes an automated cyber threat attribution method based on analyzing DNS patterns, linking them to known attack campaigns and threat actors. This research primarily contributes to Axis 2 (Analytical Domain), specifically Digital & Network Forensics, by leveraging passive DNS services and domain registration data as key network-based artifacts. It also implicitly aids Adversary Behavioral & Campaign Analysis by correlating new incidents with previously attributed attacks based on domain usage patterns. By providing a data-driven approach, this study directly aids in achieving Tactical and Operational levels within Axis 1 (Attribution Confidence & Granularity). Furthermore, its ability to identify threat actors and link cyber incidents to known campaigns contributes to classifying profiles within Axis 3 (Adversarial Motivation & Profile), strengthening targeted attribution in modern cyber threat intelligence workflows.

This framework, by facilitating a systematic and comprehensive approach to analyzing cyber incidents and research, enables a clearer understanding of the types of evidence used, the level of attribution achieved, and the specific threat actors involved. Its value is significant in research and cybersecurity practice as it helps in highlighting gaps in existing studies, ensuring a balanced and well-rounded analysis. By mapping cases and studies to these dimensions, researchers and

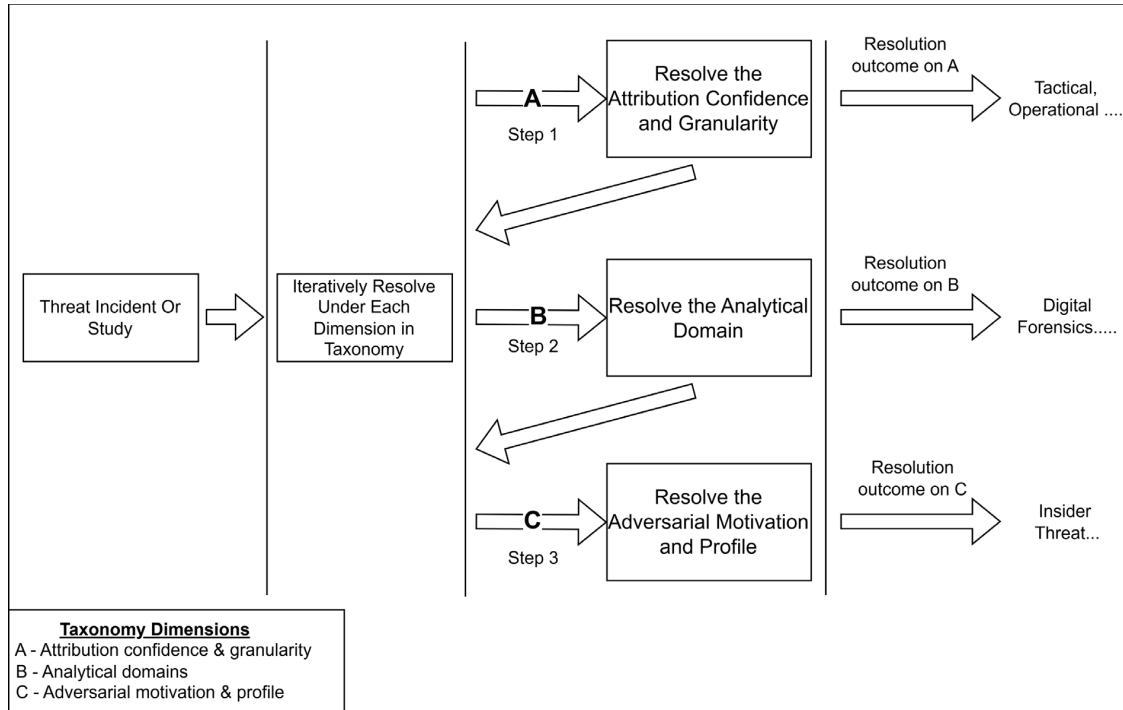


Fig. 7. Process for categorizing studies and cases on threat attribution taxonomy.

analysts can determine whether an investigation is technically-driven, strategically motivated, or actor-specific, enhancing comparability and supporting decision-making in cyber defense and policy formulation. To further streamline the attribution process and aid the practitioners, a recommended flow of the attribution process from one level to another is depicted in Fig. 8 which demonstrates the possible actions and decisions that may possibly occur during the resolution of the threat entity.

5. Cyber threat attribution data and datasets

Effective cyber threat attribution, the intricate process of identifying the responsible parties behind a cyberattack, fundamentally hinges on the meticulous collection, analysis, and synthesis of diverse data sources. These sources yield various indicators, from low-level observables like malicious file hashes to high-level abnormal system behaviors. In this process, Indicators of Compromise (IOCs) are forensic artifacts pointing to a compromise, while Tactics, Techniques, and Procedures (TTPs) describe how adversaries operate. Table 7 categorizes and describes typical features extracted from these data sources. This section delves into the types of data essential for attribution, analyzes the systemic challenges in the data ecosystem, and evaluates the capabilities and limitations of existing datasets and tools.

Attribution data can be broadly categorized into three indispensable layers: *Technical Artifacts* (e.g., network logs, malware samples), *Behavioral Patterns* (e.g., user activity, system anomalies), and *Contextual Intelligence* (e.g., threat intelligence reports, geopolitical information). Technical artifacts, forming the bedrock of forensic efforts (Lin, 2016a), offer granular insights into attack paths, adversary tools, and compromised systems. Behavioral patterns capture deviations from expected conduct, while contextual intelligence, derived from sources like threat intelligence reports, Open Source Intelligence (OSINT) from public platforms (Ruohonen et al., 2024), dark web forums, and geopolitical information, illuminates adversary motivations, broader strategic contexts, and TTPs. Each layer presents unique collection, validation, and analytical challenges, emphasizing that dataset diversity is paramount for comprehensive attribution.

5.1. Systemic challenges in attribution data ecosystems

Despite the critical necessity for diverse data, the ecosystem of attribution data is fraught with systemic challenges that profoundly hinder reliable and high-confidence attribution efforts. Understanding these complexities is crucial for developing robust attribution methodologies.

5.1.1. Data volume, velocity, and variety

The sheer volume, velocity, and variety of data generated by modern cyberattacks pose a formidable “Big Data” problem for attribution. Capturing, storing, and analyzing petabytes of network flow data or terabytes of endpoint logs in real-time or retrospectively can overwhelm even sophisticated analytical infrastructures (Munappy et al., 2019). This immense scale directly impacts the feasibility and timeliness of attribution, demanding massive computational scalability and advanced data processing capabilities that often exceed available resources. The ephemeral nature of many modern attack traces, like volatile memory artifacts or short-lived C2 communications, further compounds the challenge of timely capture and analysis. Investigators often employ techniques such as intelligent sampling, targeted data collection, and distributed processing frameworks to manage these overwhelming data streams, though these workarounds introduce their own limitations regarding completeness.

5.1.2. Data quality, veracity, and temporal relevance

Beyond sheer scale, critical limitations lie in data quality, veracity, and temporal relevance. Publicly available datasets and real-world intelligence feeds are frequently marred by data veracity challenges, including the presence of false IOCs that can be deliberately planted by adversaries in false flag operations, or generated by noisy detection systems, leading to significant misdirection. The issue of outdated samples and indicators is pervasive; the rapid evolution of malware and TTPs means that many publicly available datasets or threat intelligence indicators quickly lose their relevance, failing to reflect current adversary behaviors. This creates a “context gap”, where real-time feeds

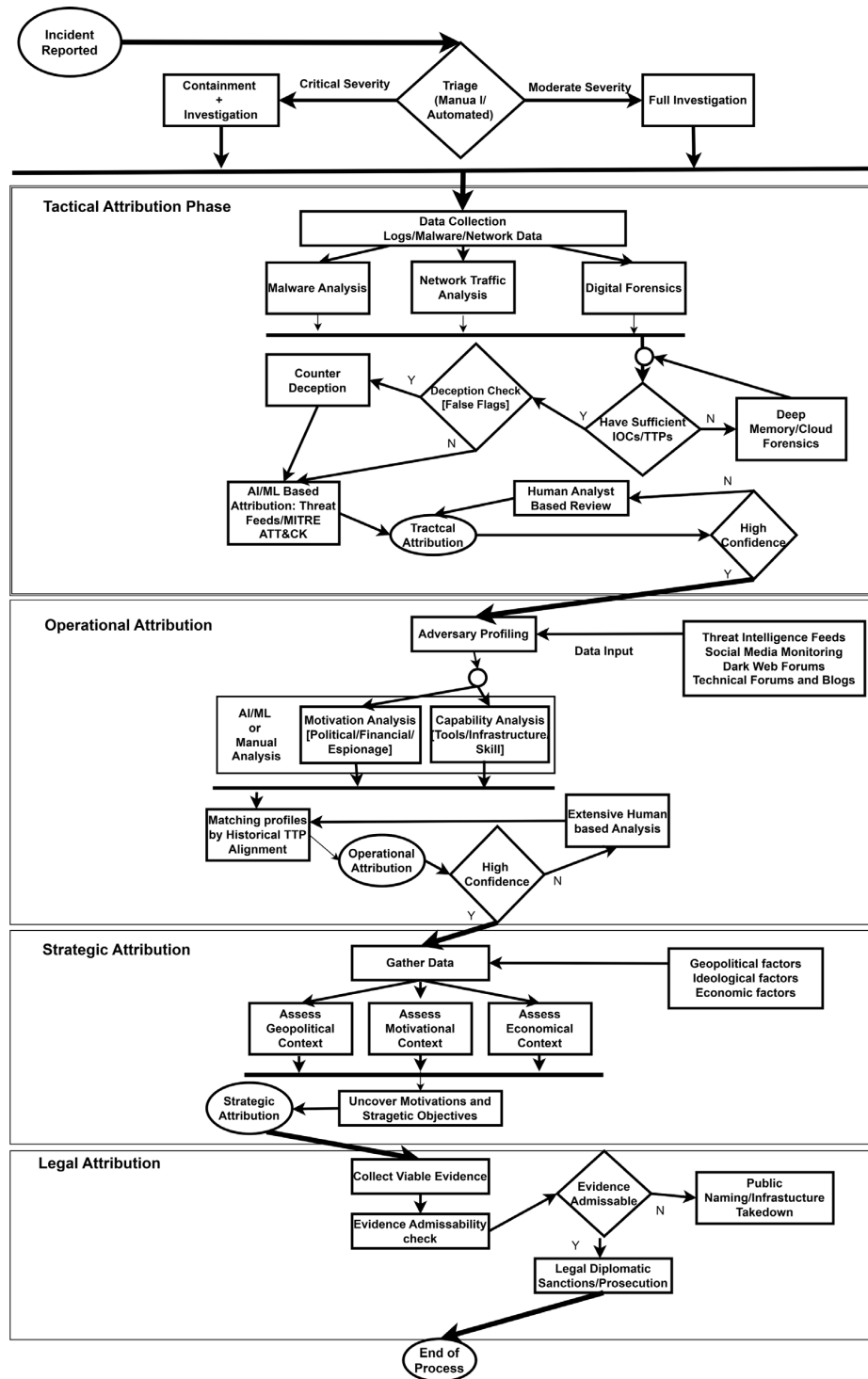


Fig. 8. Recommendation chart for practitioners.

often lack the rich contextual information (e.g., complete execution environments, pre-attack reconnaissance) necessary to accurately infer adversary intent or specific TTPs. This gap directly impacts attribution models by leading to misinterpretations of partial evidence or poor performance against novel attack patterns, requiring investigators to laboriously reconstruct missing context through additional forensic work.

5.1.3. Ground truth and dataset biases

The challenge of establishing ground truth—verifiable attacker identities linked to specific incidents—is a fundamental hurdle for developing and validating attribution models, particularly for machine learning. The scarcity of high-confidence, fully labeled real-world datasets directly impacts the reliability and generalizability of AI/ML models. Furthermore, publicly available data frequently exhibits inherent biases. For example, geographic overrepresentation of incidents in certain regions (e.g., due to higher reporting rates from well-resourced nations) can skew attribution models to disproportionately identify threat actors from those areas. Similarly, threat type bias, such as the overrepresentation of Windows malware samples, can lead to models that perform poorly against attacks targeting other operating systems or less common attack vectors, inadvertently limiting attribution capabilities to specific threat groups.

5.1.4. Interoperability and data fusion complexities

Despite the critical need for integrating heterogeneous data sources, interoperability and data fusion present significant technical and semantic challenges. Successfully correlating disparate data types—from structured network logs and malware signatures to unstructured threat reports, social media chatter, and nuanced geopolitical intelligence—requires sophisticated semantic normalization and technical fusion capabilities. The lack of universal standards for data formats, taxonomies, and exchange protocols across diverse sources makes achieving a coherent, comprehensive picture for attribution a formidable task. Examples of effective multi-source correlation often rely on manual, labor-intensive efforts by expert analysts, highlighting the need for automated semantic interoperability solutions.

5.1.5. Ethical and legal constraints in data usage

Acquiring and utilizing attribution-relevant data is deeply intertwined with ethical and legal constraints. Beyond general privacy concerns, the use of data from sources like honeypots or information acquired through clandestine means can raise significant ethical dilemmas regarding consent, data integrity, and potential entrapment. Furthermore, strict data privacy regulations (e.g., GDPR, CCPA) and data sovereignty laws introduce substantial legal hurdles for cross-border data transfer and sharing, directly impacting the collaborative investigations essential for robust attribution and creating conflicts that impede the free flow of crucial evidence. Beyond these formal regulations, practical barriers also arise: organizations may exhibit reluctance due to proprietary data concerns, fearing exposure of sensitive business operations or intellectual property. Similarly, competitive interests within industries or national security imperatives (such as protecting intelligence sources and methods) can lead to data siloing, even among entities with shared defense objectives. This complex interplay of legal compliance, commercial interests, and intelligence safeguarding often necessitates intricate negotiations and trust-building mechanisms to facilitate the necessary data exchange. Best practices for ethical data collection emphasize principles of necessity, proportionality, data minimization, and transparency. Future legal frameworks or regulatory recommendations are needed to streamline cross-border data sharing for forensic purposes while maintaining robust privacy protections, potentially through international data trusts or standardized legal gateways for data access.

5.2. Attribution datasets and tools: Capabilities and limitations

Despite the aforementioned challenges, a rich ecosystem of public datasets and tools provides foundational resources for cyber threat attribution research and operations. These resources offer diverse capabilities, but their inherent limitations necessitate critical evaluation and strategic integration for effective use.

Public Threat Intelligence Platforms: Resources like MITRE ATT&CK (MITRE Corporation, 2024a), VirusTotal (2024) and Salem et al. (2021), and the Malware Information Sharing Platform (MISP) (Wagner et al., 2016; MISP Project, 2024) provide structured threat intelligence, malware analysis, and IOC sharing. While invaluable for tactical insights, they often offer limited context on origin or may contain unverified community data.

Malware Analysis Sandboxes: Tools such as Hybrid Analysis (2024), Cuckoo Sandbox (2024a) and Cuckoo Sandbox (2024b) provide dynamic analysis of malware behavior, yielding rich forensic details. However, they can be evaded by advanced malware and may produce false positives.

Vulnerability Databases: The Common Vulnerabilities and Exposures (CVE) database (MITRE Corporation, 2024b) serves as a standardized catalog of vulnerabilities, crucial for linking attacks to specific exploits and threat groups. Its limitation lies in not providing exploit data or real-time threat context.

Network Traffic Datasets: Academic datasets like UNB ISCX (Canadian Institute for Cybersecurity (CIC), University of New Brunswick, 2012), ISOT Botnet (Information Security and Object Technology (ISOT) Research Lab, University of Victoria, 2010), CAIDA (Center for Applied Internet Data Analysis (CAIDA), 2024), MAWI (MAWI Working Group, WIDE Project, 2024), and UNIBS (Networking Research Group, University of Brescia, 2024) offer valuable network traffic for IDS research and anomaly detection. While useful for reproducible research (e.g., UNB ISCX datasets have significantly impacted IDS model benchmarking), many are static, outdated, or anonymized, limiting their direct applicability to real-time attribution.

A comprehensive overview of various attribution datasets and tools, detailing their utilization, data types, licensing, update frequency, and specific strengths and weaknesses crucial for assessing their utility in different attribution contexts, is provided in Tables 8. Effective dataset selection requires understanding these nuances, moving beyond simple availability to consider data freshness, veracity, relevance to specific threat types, and the inherent biases that might skew analytical outcomes. Common pitfalls in dataset usage include relying solely on single-source data, ignoring temporal relevance, or failing to verify data veracity through cross-referencing against multiple sources.

5.3. Future of attribution data ecosystems

The future of attribution critically depends on overcoming current data-related limitations, moving beyond traditional collection and analysis toward more intelligent, resilient, and ethically sound data ecosystems. Emerging trends and innovations offer promising avenues for more robust data acquisition and analysis, driving the next generation of attribution capabilities.

AI/ML techniques will play a transformative role, extending beyond current applications to enable more sophisticated data handling. Future research should focus on developing generative AI models capable of synthesizing realistic, anonymized training data to overcome issues of data scarcity and bias, thereby improving the generalizability of attribution models without compromising privacy. Furthermore, exploring reinforcement learning for active data collection and triage could enable attribution systems to intelligently prioritize and acquire crucial forensic artifacts from vast, noisy data streams in real-time. This includes automating the identification and filtering of false Indicators of Compromise (IOCs) and enhancing the contextualization of partial data through intelligent data fusion across heterogeneous sources.

Table 7
Representative features for cyber threat attribution.

Feature category	Feature example	Description
Temporal	Timestamps Attack lifecycle phases	Time-based indicators such as event logs, system clocks, and intrusion times. Duration and progression of attack stages (e.g., reconnaissance, exploitation).
Geospatial	IP geolocation Regional threat trends	Mapping IP addresses to physical locations to infer threat origins. Location-based threat analysis and region-specific campaign identification.
Behavioral	Malware execution patterns User activity anomalies	Dynamic malware behavior (e.g., API calls, file creation, privilege escalation). Deviations from normal user behavior such as late-night logins or lateral movement.
Network	Communication flows IP and port metadata	Inbound/outbound traffic patterns, C2 communications, data exfiltration. Technical identifiers used in transmission, including protocol, port, and routing data.
Social	Phishing indicators Social media artifacts	Traits in phishing emails or lures, like spoofed domains or urgent language. Adversary-linked profiles or campaign messaging on public platforms.
Indicators of Compromise (IOCs)	File hashes Abnormal system states	Signatures of known malware files (MD5, SHA-1, SHA-256). Unusual system behaviors, altered configurations, or untrusted processes.
Infrastructure	Domain and hosting info Attacker footprint	Metadata on domains, IPs, ASNs, and hosting providers. Reused infrastructure, digital certificates, or unique toolsets tied to threat actors.
Digital forensics	File metadata Memory artifacts	Timestamps and attributes (creation, modification, access) on local files. Volatile evidence including process injections, shellcode, or keyloggers.

Blockchain technology holds substantial promise for ensuring the integrity, immutability, and verifiable chain of custody for digital forensic evidence. Future research should investigate the design and implementation of permissioned blockchain networks for secure, auditable cross-organizational evidence sharing, addressing challenges related to scalability, data governance, and interoperability with existing forensic toolsets. Exploring the application of zero-knowledge proofs on blockchain could also enable the verification of evidence authenticity without exposing sensitive details, balancing transparency with privacy.

Furthermore, advancements in quantum computing, while still in early stages for practical application, present long-term research directions for enhancing analytical capacity. This includes investigating the theoretical application of quantum-inspired optimization algorithms for accelerating complex graph-based attribution queries over massive, multi-modal cyber threat intelligence knowledge graphs, potentially revolutionizing the processing speed and analytical capacity for truly near real-time attribution. Researchers should also continue to develop quantum-resistant cryptographic techniques to ensure the enduring integrity and confidentiality of attribution-relevant data against future quantum threats.

Continuous innovation in data collection, processing, and ethical sharing mechanisms is paramount. This necessitates interdisciplinary research into automated data labeling and enrichment pipelines for unstructured threat intelligence, and the development of privacy-preserving analytics frameworks that enable cross-jurisdictional collaboration while adhering to stringent data protection regulations. Ultimately, the goal is to build adaptive and resilient data ecosystems that can keep pace with evolving adversarial tactics and support high-confidence attribution in a dynamic cyber landscape.

6. Cyber threat attribution frameworks

Frameworks provide investigators with essential roadmaps to navigate the intricate landscape of cyberattacks. They systematically guide the identification of threat actors, help in understanding their motivations, and facilitate the development of effective countermeasures. These structured approaches are crucial for synthesizing diverse evidence and supporting complex attribution decisions.

6.1. Components of threat attribution frameworks

Attribution frameworks offer a conceptual structure for systematic cyber defense, while attribution models operate within them to interpret evidence and generate conclusions. Effective attribution frameworks are built upon several critical, interconnected components that

collectively enable a comprehensive understanding of cyber incidents and their perpetrators.

Evidence Collection and Analysis forms the foundational layer of attribution. It involves the meticulous gathering of digital artifacts from myriad sources to construct a cohesive picture of a cyberattack. This encompasses network logs (providing historical communication patterns [Ren et al., 2022](#)), system forensics (uncovering traces of intrusion and malware), and malware samples (offering insights into attacker tools and infrastructure). Open-source intelligence (OSINT) further expands this by collecting publicly available information to contextualize attacks and reveal broader patterns ([Nash, 2021](#)). While foundational, this stage faces significant challenges from the sheer volume and complexity of data, as well as adversary obfuscation. Emerging technologies like AI and machine learning enhance this process by automating analysis and detecting subtle connections ([Mohurle and Patil, 2017](#); [Akbanov et al., 2019](#)). However, the accuracy of AI-based systems remains highly dependent on the quality and representativeness of training data. For instance, ML classifiers trained predominantly on Windows malware may fail against Linux-based threats or advanced nation-state toolkits, as adversarial evasion or dataset bias can significantly mislead AI ([Guembe et al., 2022](#)). These challenges highlight the importance of integrating diverse evidence sources to strengthen attribution.

Attribution Models serve as the structured methodologies guiding analysts through the complex process of interpreting evidence. These models synthesize collected data to provide a comprehensive approach for understanding not only the technical aspects of an attack, but also the motivations, capabilities, and infrastructure involved. They incorporate multi-dimensional approaches to analyzing Tactics, Techniques, and Procedures (TTPs), forming behavioral signatures linkable to known threat actors ([Bierwirth et al., 2024](#); [Ahmed et al., 2021](#)). For example, attribution models proved crucial in the SolarWinds breach, connecting supply chain exploitation TTPs to sophisticated state-sponsored actors and analyzing infrastructure to attribute with high confidence ([Anisa and Widianingsih, 2024](#)). Despite their utility, these models face limitations as attackers constantly alter behavior, and new techniques like deepfakes and AI-driven malware challenge traditional paradigms. The “black-box” nature of AI/ML also poses risks of bias and misinterpretation ([Guembe et al., 2022](#)), yet these models remain vital for uncovering broader operational and strategic contexts.

Attribution Levels define the varying degrees of insight into a cyberattack’s nature and origin, from immediate technical identification to strategic understanding and legally admissible evidence. These hierarchical levels (Tactical, Operational, Strategic, Legal), as comprehensively detailed in our IAM Axis 1 (Section 4-A), are a core conceptual

Table 8

Comprehensive overview of cyber threat attribution datasets and tools, including capabilities and limitations.

Dataset/Tool	Utilization	Data type	Access	Managed by	Update Freq.	Acad-emic use	API	Strengths	Weaknesses
APTnotes	APT group tracking, context enrichment	Threat reports (PDF, HTML)	Public	Community	Irregular	Yes	No	Aggregates public APT documentation from diverse sources	Unstructured; requires manual parsing
OTX	IOC sharing, TTP tracking	IOCs	Freemium	AlienVault (AT&T)	Real-time	Yes	Yes	Broad user base, real-time data, community enriched	Community data may lack verification
ThreatFox	IOC and C2 enrichment	IOC feeds	Public	Abuse.ch	Frequent	Yes	Yes	Actionable C2 and malware indicators	Low behavioral/contextual depth
Malpedia	Malware attribution, actor mapping	Malware metadata, YARA	Free (login)	Fraunhofer FKIE	Frequent	Yes	Limit-ed	Curated YARA rules, behavior summaries	Windows-centric, limited OS diversity
Any.run	Interactive malware sandboxing	Sandbox logs	Freemium	Any.run	Real-time	Yes	Yes	Visual, interactive analysis	Limited free access, evasive malware issues
MITRE ATT&CK	TTP mapping, red teaming	Structured TTPs	Public	MITRE	Regular	Yes	Yes	Authoritative, widely adopted taxonomy	Descriptive only; no raw data
VirusTotal	IOC enrichment, malware detection	Executables, URLs, AV results	Freemium	Google (Chronicle)	Real-time	Yes	Yes	Large-scale AV aggregation; broad IOC coverage	Limited provenance/context; rate-limited API
MISP	Threat intel sharing, IOC correlation	STIX-like structured data	Open-source	CIRCL (LU)	Commun-ity driven	Yes	Yes	Collaborative, widely adopted platform	Contributor-dependent data quality
Hybrid Analysis	Dynamic malware analysis	Behavior reports	Freemium	CrowdStrike	Frequent	Yes	Yes	Behavior-rich forensic reports; dynamic sandboxing	False positives; can be evaded
Cuckoo Sandbox	Local malware analysis	Dynamic sandbox logs	Open-source	Community	Variable	Yes	Yes	Fully customizable, offline analysis	Requires setup; limited evasion handling
CVE	Vulnerability tracking	Structured CVE records	Public	MITRE	Daily	Yes	Yes	Globally accepted vulnerability taxonomy	Lacks exploit code or live threat context
UNB ISCX/CIC	ML model training, traffic analysis	Labeled Net Traffic (PCAP, CSV)	Public	Univ. of New Brunswick	Static	Yes	No	Standardized, ML-ready datasets	Some outdated scenarios; synthetic in parts
ISOT Botnet	Botnet analysis	Botnet PCAPs	Public	Univ. of Victoria	Static	Yes	No	Real botnet traffic samples	Botnet-specific only; dated scenarios
CAIDA	Internet-scale traffic analytics	Raw & anonymized NetFlow	Public (request)	UC San Diego	Regular	Yes	No	Huge scale; useful for infra-level attacks	Lacks app-level context; anonymized
MAWI	ISP-level traffic trends	Anonymized ISP traffic	Public	WIDE Project	Daily	Yes	No	Continuous updates; good for anomaly detection	No attack labels; noisy
UNIBS	Attack detection, traffic modeling	Labeled traffic flows	Public (request)	Univ. of Brescia	Static	Yes	No	Focused attack traces	Limited volume; not up-to-date
Leonard Bradatsch Dataset	Anomaly detection, IDS benchmarking	PCAPs, NetFlows, Argus logs	Mixed	Univ. of Ulm	Static	Yes	Partial	Rich features across formats	Limited accessibility; varied licenses

component within frameworks, guiding the depth and purpose of the attribution effort.

Confidence Levels play a vital role in communicating the certainty associated with any given attribution. Cyber threat attribution is rarely definitive; it involves inherent uncertainty that frameworks must transparently acknowledge. Confidence levels indicate the strength of evidence and likelihood of accuracy, with high confidence implying robust, multi-source evidence and low confidence suggesting speculative conclusions. Some frameworks adopt structured scales (e.g., ICD 203's confidence modifiers: low, moderate, high) to formalize assessments. As seen with the WannaCry attribution, confidence levels evolve with accumulating evidence (Colbaugh and Glass, 2011; Anisetti et al., 2020; Morag and Maouda, 2021). Explicitly stating confidence levels helps

decision-makers calibrate responses, especially for high-stakes attributions involving nation-states, where misattribution can lead to severe geopolitical consequences (Hettema, 2021). Transparent communication fosters intellectual honesty and manages inherent uncertainties.

These components – evidence collection, attribution models, levels, and confidence – are deeply interconnected. Together, they shape a holistic understanding of cyberattacks and must evolve alongside threats. Emerging technologies such as AI and ML provide new tools for attribution, but they introduce challenges such as the risk of bias and the need for transparency. By integrating technical analysis, behavioral profiling, and geopolitical insights, attribution frameworks can empower organizations and governments with the intelligence needed to stay ahead of cyber threats. Ultimately, the goal is to move from

reactive to proactive attribution, enabling defenders to anticipate future attacks and mitigate risks before they materialize.

6.2. Critical analysis of existing threat attribution frameworks

Several prominent frameworks are widely utilized in cyber threat attribution, each offering distinct strengths and perspectives. However, a critical analysis reveals their individual limitations, particularly when confronted with the escalating complexity of modern cyber warfare. Table 9 provides a comparative overview of these common frameworks, highlighting their fundamental aspects, strengths, and weaknesses.

The Diamond Model of Intrusion Analysis offers a comprehensive, interdisciplinary approach by examining cyberattacks through the dynamic interplay of four core components: adversary, capability, infrastructure, and victim (Caltagirone et al., 2013). Its primary strength lies in emphasizing the complex relationships and dependencies among these elements, facilitating collaborative analysis and providing a holistic view of the intrusion. This model has proven valuable in dissecting sophisticated campaigns, such as certain APT28 operations. However, while providing a holistic view, the Diamond Model's lack of prescriptive steps for evidence integration limits its direct utility as a granular attribution methodology. Its abstract nature also struggles to guide real-time, multi-phase attribution for highly adaptive campaigns, where the precise sequence of adversary actions or evolving capabilities demand granular, actionable insights not explicitly provided by its high-level components.

The Cyber Kill Chain, developed by Lockheed Martin, provides a linear, sequential framework for understanding the phases of a cyber-attack, from reconnaissance to data exfiltration (Tarnowski, 2017). Its intuitive simplicity makes it widely accessible for incident response teams, offering clear opportunities to detect and disrupt attacks at various stages. Despite its widespread adoption, its strict linearity fails to capture the iterative or non-sequential nature of modern APT operations, where adversaries may loop back, bypass, or combine stages (e.g., direct exploitation without traditional weaponization). More critically, its overwhelming focus on the technical attack lifecycle often omits crucial geopolitical, economic, or psychological factors that drive adversary motivations, providing limited context for why an attack occurred and thus hindering strategic attribution to actors beyond technical indicators.

The MITRE ATT&CK Framework is a comprehensive and continuously evolving knowledge base that catalogs adversary tactics, techniques, and procedures (TTPs) across the entire attack lifecycle (Akbar et al., 2022). It is highly effective at providing detailed, real-world observed attacker behaviors, enabling organizations to map specific methods and tailor defensive measures. This granular insight is invaluable for proactive threat hunting and defensive planning, as demonstrated by its application in understanding complex, multi-stage incidents like the SolarWinds attack (Abo El Rob et al., 2024). Yet, as a descriptive knowledge base, MITRE ATT&CK does not provide a prescriptive methodology for integrating disparate evidence points into a cohesive attribution conclusion. While invaluable for mapping techniques, its reliance on observed behaviors means it struggles to keep pace with rapidly evolving zero-day exploits or novel TTPs from agile threat actors, requiring constant manual updates that limit its proactive attribution capabilities against entirely new attack patterns.

While structured and logical, the Pyramid Model of Attribution categorizes attribution into three hierarchical levels: technical, organizational, and political (Tsagourias and Farrell, 2020). This approach offers a clear progression from identifying tools to identifying the sponsoring entity, providing a framework for assigning accountability. However, the Pyramid Model often oversimplifies the messy reality of modern cyber attribution, where evidence for technical, organizational, and political levels may not neatly align or be consistently available. Its rigid hierarchy particularly struggles with deceptive operations, like

false flags, where technical indicators (e.g., infrastructure) are deliberately manipulated to point away from the true perpetrator, making it difficult to ascend confidently to higher attribution levels without comprehensive cross-referencing and contextual intelligence.

The Evidence-Based Attribution Model (EBAM) emphasizes the rigorous collection and analysis of verifiable evidence to support attribution claims, thereby minimizing the risk of false attribution (Čerget and Hudec, 2023). Its core strength lies in promoting accountability and transparency by demanding substantiated evidence, making attributions credible and legally defensible, particularly in legal and policy contexts. Yet, EBAM's meticulous, human-intensive nature renders it prohibitively slow and resource-intensive for rapid incident response or real-time attribution. The stringent evidentiary thresholds, while crucial for legal contexts, can impede agile investigations by demanding a level of proof that is often unattainable in the ambiguous, ephemeral landscape of cyber operations, particularly when dealing with sophisticated, anti-forensic techniques.

Beyond these widely recognized models, various other frameworks offer unique perspectives on cyber threat attribution. The Unified Kill Chain combines the strengths of the Cyber Kill Chain and MITRE ATT&CK, providing a more comprehensive view of the attack lifecycle (Matto, 2024). The Intelligence Cycle offers an iterative process for refining attribution hypotheses through continuous collection, analysis, and dissemination of information. Incident response frameworks like NIST's Computer Security Incident Handling Guide and SANS' Incident Handler's Handbook, while not explicitly designed for attribution, offer valuable guidelines for evidence collection, analysis, and reporting, thus indirectly aiding attribution efforts. Yet, in isolation, each of these frameworks falls short of capturing the multidimensional complexity of real-world attribution.

6.3. Synthesis: The imperative for integrated and adaptive frameworks with AI/ML integration

The limitations of individual frameworks necessitate a synthesis that integrates their strengths while compensating for individual weaknesses. The true value of these frameworks lies in their complementary and combined use to overcome inherent limitations and exploit individual strengths. Rather than limiting to a single framework, a multi-model approach enables a more robust attribution process. For instance, initial attack mapping might utilize the Cyber Kill Chain, followed by MITRE ATT&CK for detailed behavioral analysis. The Diamond Model could then contextualize the adversary's relationships and capabilities, with Graph Neural Networks (GNNs) significantly improving its 'Infrastructure' component by automatically discovering hidden connections and identifying shared attacker resources across vast datasets. The Evidence-Based Attribution Model (EBAM) would then provide systematic validation of evidence, potentially enhanced by Explainable AI (XAI) techniques to ensure the interpretability of complex findings for legal defensibility. This collective use helps compensate for missing aspects encountered when selecting only a single framework, as demonstrated by real-world incidents like SolarWinds, where combining different analytical perspectives was vital for developing a full attribution narrative.

As the threat landscape evolves, so too must attribution frameworks. Artificial Intelligence and Machine Learning offer transformative capabilities to augment traditional models. The following roadmap outlines key areas for integrating AI/ML into existing attribution frameworks:

To enhance the Diamond Model with Graph AI, there is a need for more advanced approaches. Leveraging Heterogeneous Graph Neural Networks (HGNNs) can automatically map and cluster complex relationships between adversaries, their capabilities (like malware types or exploit kits), and their shared infrastructure (such as C2 servers, domains, and IPs) (Lindsay, 2015; Farooq and Otaibi, 2018). GNNs are adept at uncovering hidden connections and identifying patterns

that link disparate attacks to a common actor group, thereby refining the ‘Adversary-Infrastructure-Capability’ relationships (Idika and Mathur, 2007; Chester and Allenby, 2020). Furthermore, employing advanced behavioral analytics and clustering algorithms helps identify consistent TTPs across incidents, linking fragmented attacks to specific adversary groups based on their unique operational fingerprints within the model’s ‘Capability’ and ‘Adversary’ components (Sun et al., 2023; Irshad and Siddiqui, 2024; Rawat et al., 2021).

For augmenting the Cyber Kill Chain with Predictive AI, integration of Machine Learning for anomaly detection at each stage is crucial (Nakhodchi et al., 2021; Bilen and Özer, 2021). This means identifying anomalous reconnaissance traffic and predicting weaponization success rates, allowing predictive analytics to anticipate adversary next steps and inform proactive disruption measures. Natural Language Processing (NLP) can also automate the extraction of threat intelligence from unstructured open sources, such as dark web forums and threat reports. This enriches early reconnaissance and weaponization phases, feeding contextual data directly into detection systems (Noor et al., 2019; Wang et al., 2021; Ren et al., 2022).

Strengthening MITRE ATT&CK with Adaptive AI requires a multi-pronged approach. Utilizing Large Language Models (LLMs) and Natural Language Processing (NLP) for automated mapping of unstructured threat intelligence reports and incident logs directly to specific ATT&CK techniques can provide real-time enrichment and ensure the knowledge base remains current (Zhang et al., 2024; Böge et al., 2025). For identifying novel TTPs, Few-shot learning can help classify new attack patterns that have not been explicitly cataloged, allowing the framework to adapt more quickly to emerging threats without extensive retraining (Omar et al., 2024; Chen et al., 2024). Additionally, integrating Reinforcement Learning models that learn from observed adversary TTPs can predict likely follow-on actions or preferred evasion techniques, enhancing defensive countermeasures aligned with specific ATT&CK tactics (Basnet et al., 2024; Oh et al., 2024).

To improve the Pyramid Model with Probabilistic AI, it is essential to employ Probabilistic Graphical Models. These can integrate diverse technical, organizational, and geopolitical evidence, allowing for the calculation of confidence scores at each attribution level (Bendiek and Schulze, 2021). For transparent justification, Explainable AI (XAI) techniques can provide clear reasons for AI-driven attribution decisions, which is crucial for building trust and achieving higher-level (strategic/political) consensus, especially when dealing with ambiguous or contradictory evidence like false flags (Skopik and Pahi, 2020; Mehrabi et al., 2022; Hassija et al., 2024).

Finally, accelerating EBAM with Automation & Verification involves automating the initial stages of evidence collection and classification using Machine Learning classifiers to process vast volumes of data, including malware samples, network logs, and unstructured text (Mohurle and Patil, 2017; Akbanov et al., 2019). NLP can rapidly categorize threat intelligence reports, while Computer Vision might analyze visual artifacts. This significantly reduces manual overhead, speeding up the rigorous EBAM process. Integrating Blockchain technology ensures an immutable and verifiable chain of custody for digital artifacts, strengthening evidentiary standards even with automated collection processes (Chatziamanetoglou and Rantos, 2024; Gunawardena, 2023; Glady et al., 2024).

The rise of AI-driven attacks, including deepfakes and advanced AI-enabled obfuscation (Gupta et al., 2023; Takale et al., 2024; Teichmann, 2023; Guembe et al., 2022), further highlights the critical need for frameworks to dynamically adapt and integrate these capabilities. This necessitates not just the use of AI/ML tools within existing frameworks, but a deeper re-conceptualization of the frameworks themselves.

7. Data-driven approaches

The escalating sophistication and volume of cyberattacks demand a fundamental transformation in attribution methodologies. While traditional methods have served their purpose, their limitations in addressing the complexity and dynamism of modern threats necessitate innovative, data-driven techniques.

7.1. Traditional techniques

Traditional approaches, such as signature-based detection and heuristic analysis, have long been foundational in cybersecurity. Signature-based methods rely on predefined patterns, like unique malware hashes, for identification (van Ovayan et al., 2024). Heuristic analysis, on the other hand, attempts to identify threats based on rule-based behavioral patterns, even for unknown samples (McGarret et al., 2024). While these methods are effective against known threats, they are inherently reactive. They struggle against novel and polymorphic threats that constantly change to evade detection, leading to high false positive rates against sophisticated adversaries employing advanced evasion techniques. The widespread use of anonymization, obfuscation, and false flag tactics by attackers further severely impedes the analysis of forensic artifacts like network logs and system configurations. Critically, these traditional methods lack the scalability and adaptability required for modern threats, demanding continuous manual updates and significant expert intervention (Idika and Mathur, 2007; Chester and Allenby, 2020). This starkly contrasts with AI-driven approaches, which learn and adapt automatically, enabling a more proactive and resilient attribution capability.

7.2. Machine learning enabled attribution

Machine learning (ML) and artificial intelligence (AI) are fundamentally transforming cyber threat attribution by enabling the analysis of vast datasets with unparalleled speed and precision. These technologies uncover subtle patterns and connections that human analysts might miss, learning from historical attack data to anticipate future attack vectors and attribute them with greater accuracy. This versatility empowers organizations to respond proactively. For instance, Heterogeneous Graph Neural Networks (HGNNs) are adept at modeling complex relationships between diverse entities in an attack, like attackers, malware, and infrastructure (Lindsay, 2015; Farooq and Otaibi, 2018). Dynamic Temporal Graph Neural Networks (DT-GNNs) further enhance this by tracking how these relationships evolve over time (Jha et al., 2021). Federated learning offers a privacy-preserving solution, allowing collaborative model training across organizations without direct data sharing (Ammara et al., 2024; Sikos, 2020). Adversarial machine learning fortifies attribution models by simulating attack scenarios, making them more resistant to evasion (Zhu et al., 2024; Dixit, 2024). Meanwhile, Explainable AI (XAI) addresses the “black box” problem, providing human-understandable explanations for AI-driven decisions, fostering trust and transparency (Hassija et al., 2024). Zero-shot learning offers the tantalizing prospect of rapidly attributing new, unseen attacks by generalizing from past data (Omar et al., 2024; Chen et al., 2024).

7.3. Emerging techniques

Beyond general ML applications, several specialized AI-driven techniques are emerging, offering powerful new lenses for attribution as shown in Fig. 9.

Graph-Based Techniques: By representing cyberattack entities and their relationships as interconnected nodes and edges, graph-based techniques offer a powerful visualization and analytical framework (Idika and Mathur, 2007; Chester and Allenby, 2020). This allows for the discovery of hidden connections, such as coordinated activities or

Table 9
Comparison of threat attribution frameworks.

Aspect	Diamond model of intrusion analysis	The cyber kill chain	MITRE ATT&CK framework	Evidence-Based Attribution Model (EBAM)
Purpose	Focuses on understanding the relationships between adversary, infrastructure, capability, and victim.	Models the stages of a cyber attack to disrupt or prevent adversarial actions.	Catalogs adversary behaviors and techniques to understand how they attack and how to defend.	Provides a systematic approach to attributing cyber attacks based on evidence.
Key components	Adversary, Infrastructure, Capability, Victim.	Reconnaissance, Weaponization, Delivery, Exploitation, Installation, Command & Control, Actions on Objectives.	Tactics, Techniques, Procedures (TTPs).	Evidence collection, Evidence classification, Attribution decision-making.
Primary focus	Understanding and analyzing relationships in attacks.	Identifying and stopping attacks at different stages.	Identifying and mitigating specific techniques used by adversaries.	Attributing responsibility for cyber attacks based on collected evidence.
Strengths	- Emphasizes relationships and dependencies among key elements. - Facilitates collaboration across organizations. - Provides a holistic view of the intrusion.	- Simple, linear model. - Widely adopted by security teams for incident response.	- Detailed and comprehensive. - Continuously updated with real-world data.	- Systematic and methodical approach. - Focus on evidence quality and context.
Weaknesses	- Can be complex and abstract. - Requires a high level of expertise. - Requires significant data to fully utilize.	- Linear model may oversimplify complex attacks. - Focuses primarily on attack lifecycle, not on attribution or adversary analysis.	- Can be overwhelming due to its detailed nature. - Attribution is not its primary goal.	- Time-consuming and resource-intensive. - Requires extensive evidence collection and analysis.
Use cases	- Advanced threat analysis. - Incident response coordination. - Can be used in combination with other models for better attribution.	- Incident response. - Threat detection and mitigation. - Can complement other models like MITRE ATT&CK for broader coverage.	- Threat intelligence. - Defensive planning and threat hunting. - Often combined with models like the cyber kill chain or Diamond Model for full attribution.	- Formal attribution of cyber attacks. - Legal and policy contexts. - Works well with MITRE ATT&CK or Diamond Model to strengthen attribution accuracy.
Application in attribution	- Analyzes connections between elements to identify patterns that may lead to attribution. - Often combined with EBAM or MITRE ATT&CK for a full picture.	- Identifies where in the attack process attribution efforts should focus. - Can be complemented with frameworks like EBAM for final attribution.	- Uses specific TTPs to link attacks to known adversaries. - Best used in combination with other models for comprehensive attribution.	- Provides a structured process for determining attribution based on evidence. - Can be strengthened by integrating TTPs from MITRE ATT&CK.
Attribution capability	High; focuses on connecting actors, tools, and targets to understand the adversary's identity.	Limited; focuses more on understanding the attack lifecycle than on attribution.	Moderate; can assist in attribution by mapping techniques to known adversary behaviors.	High; designed specifically to attribute attacks to specific actors.
Sample approach	Relating infrastructure used in multiple attacks to a single adversary.	Stopping an attack by recognizing and intervening during the delivery phase.	Mapping an observed technique to a specific known threat actor.	Collecting digital forensics and linking it to a threat group based on evidence quality.

shared Tactics, Techniques, and Procedures (TTPs) through community detection algorithms (Hiremath et al., 2023; Lee et al., 2019). Knowledge graphs take this further, providing a structured framework to map complex relationships between attackers, infrastructure, and techniques, enabling deeper analysis and inference (Iorga et al., 2020; Pastor-Galindo et al., 2020).

Natural Language Processing (NLP) Approaches: NLP is increasingly vital for extracting insights from vast amounts of unstructured textual data, including technical reports, news, social media, and dark web forums (Iorga et al., 2020; Pastor-Galindo et al., 2020). Transformer-based models like BERT and GPT excel at understanding context and nuance, identifying linguistic cues that link attacks to specific actors. Event extraction and temporal reasoning techniques reconstruct attack timelines, while zero-shot learning allows models to generalize to new attack patterns without explicit labels (Omar et al., 2024; Chen et al., 2024). NLP enriches contextual intelligence, revealing motivations and strategies often missed by purely technical analysis (Wang et al., 2021).

Simulation and Behavioral Techniques: These methods enable a proactive approach by modeling attack scenarios to understand adversary

psychology and tactics (Greitzer and Hohimer, 2011; Hacks et al., 2021). Threat actor emulation using reinforcement learning simulates adversary behavior in virtual environments, helping analysts predict their next moves (Basnet et al., 2024; Oh et al., 2024). Digital twins of network infrastructures provide safe platforms for studying real-time attack impacts and origins. Adversarial text generation aids in detecting and attributing disinformation campaigns by simulating malicious communication patterns (Allison et al., 2023; Stiff and Johansson, 2022). These techniques empower defenders to stay ahead of attackers by understanding their likely actions.

Cryptography and Blockchain-based Techniques: In an environment where evidence integrity is paramount, these technologies safeguard attribution data. Blockchain, with its decentralized and tamper-proof nature, creates an immutable ledger for secure evidence storage and sharing, fostering trust among stakeholders (Chatziamanetoglou and Rantos, 2024; Gunawardena, 2023; Glady et al., 2024). Quantum-safe cryptographic techniques address the looming threat of quantum computers, protecting attribution data against future decryption attempts

and ensuring its long-term viability (Gunawardena, 2023; Glady et al., 2024).

Hybrid and Cross-Layer Techniques: Acknowledging the multi-faceted nature of modern attacks, these approaches integrate information from various system layers, including hardware, networks, and applications. Cross-layer cyber threat attribution gathers data from diverse layers to trace sophisticated attacks from initial entry to ultimate objective, revealing comprehensive TTPs (Fathi and naeim Sobhani, 2023; Xiao et al., 2024). Cyber threat intelligence fusion platforms further enhance this by aggregating data from commercial feeds, OSINT, and social media, enabling analysts to connect disparate events and identify hidden patterns (Sun et al., 2021). These holistic techniques provide a more complete picture of the threat landscape.

7.4. Ethical and practical considerations in AI/ML attribution

The increasing reliance on AI and Machine Learning for cyber threat attribution, while offering transformative capabilities, introduces significant ethical and practical considerations. These are rooted in the potential biases within training datasets that can lead to inaccurate information, incorrect attribution, and unjust targeting of individuals or attack types (Mehrabani et al., 2022; Alam et al., 2024; Oyeniran et al., 2022). Such biases manifest in various forms: geopolitical bias (overrepresentation of actors from particular regions), technological bias (over-attention to malware-based attacks over insider threats), historical bias (models predicting based on past trends rather than emerging techniques), and class/economic bias (models trained on data from wealthier nations overlooking threats from under-resourced regions) (Mehrabani et al., 2022; Mavrogiorgos et al., 2024). These biases can exacerbate geopolitical tensions, result in false accusations, and erode trust in AI-powered attribution systems. Algorithmic bias, stemming from design choices or optimization objectives, can perpetuate unfair outcomes (Kim and Cho, 2022; Blanzeisky and Cunningham, 2021). To mitigate these, practitioners must use diverse, balanced datasets, leverage synthetic data to fill gaps (Blanzeisky and Cunningham, 2021; Shahul Hameed et al., 2024), and employ fairness algorithms for equitable results (Pagano et al., 2023; Alelyani, 2021). Routine testing and evaluation of algorithms, along with diverse stakeholder involvement, are paramount for ensuring fair and inclusive models (Handley et al., 2001; van Giffen et al., 2022; Donald et al., 2024).

Furthermore, deep learning algorithms often act as “black boxes”, generating outputs without explicit decision-making statements. This lack of transparency can make AI-powered attributions suspicious in sensitive legal or geopolitical settings, challenging accountability. Explainable AI (XAI) techniques are crucial here, providing concise, fact-related explanations for produced outputs, fostering understanding and trust in AI’s judgments (Hassija et al., 2024). Misattribution, whether from deliberate manipulation (e.g., false flags), skewed data, or technical faults, is a critical concern, as AI can amplify such errors. Misattribution leads to severe consequences, including harming innocent parties and escalating conflicts. Integration of AI analysis with human review, cross-validation with alternative sources, and confidence scoring are critical to minimizing this risk. Access to sensitive information for training AI algorithms also raises privacy concerns, which can be mitigated by methodologies like federated learning, differential privacy, and anonymization (Skopik and Pahi, 2020; Stejskal and Faix, 2022). Accountability for errors in AI/ML attribution decisions can be problematic, potentially falling on developers, users, or deploying organizations. This ambiguity can destroy trust, especially in high-consequence national security or judicial cases. Establishing clear roles, transparency in AI development, and robust accountability frameworks are essential (Zhang, 2024; Yapar, 2023). ML models are also vulnerable to adversarial manipulation; attackers introduce deceptive data to mislead detection. Adversarial training enhances model resilience by

training on adversarial examples, helping models recognize and neutralize deceptive techniques (Zhu et al., 2024; Dixit, 2024). Generative adversarial networks (GANs) simulate sophisticated threats, improving anomaly detection and forcing AI systems to adapt to dynamic threats. The deliberate planting of false evidence further complicates matters, as AI might unintentionally validate it. AI systems must be trained to identify evidence inconsistencies and incorporate comprehensive geopolitical and contextual analysis (Skopik and Pahi, 2020; Dotter et al., 2021). These measures allow AI attribution systems to be used more ethically, with increased dependability and transparency, less misuse, and more accurate and accountable attribution in cyberspace.

7.5. Performance analysis and current gaps

Recent advancements in cyber threat attribution increasingly leverage machine learning (ML), artificial intelligence (AI), and novel modeling techniques to improve attribution accuracy and scalability. This section critically evaluates key contributions, identifies persistent challenges, and outlines future research directions, supported by qualitative insights in Table 10 and quantitative comparisons in Table 11. Several studies have demonstrated the potential of ML for extracting features from unstructured Cyber Threat Intelligence (CTI) and predicting threat actor attribution. Irshad and Siddiqui (2023) introduced Attack2Vec, achieving high attribution accuracy, while Noor et al. (2019) employed semantic search (LSA) and high-level Indicators of Compromise (IOCs) to benchmark effectiveness. However, both works acknowledged issues such as dataset imbalance and limited automation (Ren et al., 2022). Basnet et al. (2024) also applied Deep Reinforcement Learning using malware and behavioral data, showing promise in attribution tasks.

Beyond standard ML, researchers have explored hybrid and behavior-aware models. Skopik and Pahi (2020) addressed false flag operations using CAM and Q-Models, integrating technical and socio-political evidence. Irshad and Siddiqui (2024) combined behavioral and technical attributes using a Threat Actor Encyclopedia dataset, while Leite et al. (2024) proposed DNS-based attribution methods, highlighting the potential of underutilized data sources despite their short-lived nature. These approaches face limitations including feature extraction complexity and limited data availability (Warikoo, 2021).

Recent studies have also explored probabilistic and modular frameworks. Teuwen (2023) proposed a framework integrating opinion pooling across modules (e.g., text analysis, malware clustering), promoting extensibility but dependent on module reliability. Wang et al. (2022) modeled malware behavior using time-series shapelets, enabling interpretable, lightweight attribution for zero-day threats, albeit limited by log quality. Zhang et al. (2024) employed Large Language Models (LLMs) to auto-populate a hierarchical attribution framework, bridging conceptual models and operational data, though lacking empirical validation.

Graph-based methods are gaining traction. Chandra et al. (2023) built a knowledge graph pipeline combining semantic modeling and DL for APT attribution, while Ren et al. (2022) created a bilingual, ontology-driven cybersecurity graph achieving strong F1 scores, albeit with real-time complexity issues. Duan et al. (2024) framed attribution as a link prediction task using STIX2-based heterogeneous graphs, capturing semantic relations but lacking evaluation under deceptive scenarios.

Additional studies contribute niche yet critical insights. Rani et al. (2024) emphasized feature selection and behavioral correlation in malware-based attribution for low-resource contexts. Lee and Choi (2025) Stress-tested classifiers using the TTP synonym mutation, revealing weaknesses in the handling of semantic obfuscation. A hybrid CNN-LSTM model by Böge et al. (2025) processed CTI text effectively but remains untested against linguistic camouflage. Edie et al. (2023) used association rule mining and activity-attack graphs from MITRE ATT&CK for APT group attribution, with strong precision but limited recall assessment.

Table 10

Key studies in cyber threat attribution: Contributions and limitations.

Study	Contribution	Limitation
Skopik and Pahi (2020)	Proposes the Cyber Attribution Model (CAM) to evaluate artifact trustworthiness and analyze false flag operations, identifying key attribution artifacts.	Critiqued for subjective artifact trustworthiness and limited automation, struggling with data integration and scalability. It does not fully address evolving attack techniques, legal/political complexities, or human factors, and lacks empirical validation.
Irshad and Siddiqui (2023)	Develops the Attack2Vec model and an ML-based attribution framework, featuring a comprehensive feature set validated against benchmarks.	Suffers from limited and imbalanced datasets, potentially missing behavioral features, and raising data privacy concerns. Lacks fully automated, real-time attribution mechanisms.
Noor et al. (2019)	Presents an ML framework using LSA-based semantic search for high-level IOCs from unstructured CTI reports, evaluating multiple models and identifying IOC overlaps.	Faces challenges with ambiguous/incomplete data and feature selection. Overlapping high-level IOCs and limited generalizability reduce accuracy.
Warikoo (2021)	Introduces the novel triangle model for improved attribution accuracy, using high-fidelity Sector, Tools, and TTPs indicators within a multi-stage analytical process integrated with MITRE ATT&CK.	Limited to specific sectors and reliant on tools vulnerable to obfuscation. Its manual process is labor-intensive, and it uses a simplified confidence scale.
Irshad and Siddiqui (2024)	Incorporates behavioral attributes and a hybrid feature set with optimal selection for high-performance empirical validation using the Threat Actor Encyclopedia dataset.	Highlights the scarcity of comprehensive behavioral datasets. Needs standardization of report formats and more sophisticated feature extraction.
Leite et al. (2024)	Proposes an automated DNS pattern-based threat attribution method using clustering and similarity scoring, providing insights into temporal DNS behavior for security analysts.	Limitations include the short lifespan of DNS patterns and reliance on specific datasets/third-party services.
Ren et al. (2022)	Develops CSKG4APT, a scalable, bilingual knowledge graph platform for APT attribution, featuring advanced entity extraction, dynamic updates, and enhanced threat hunting capabilities based on real-world evaluation.	Limitations include data timeliness, semantic gaps between low-level IOCs and high-level views, and challenges in automated, large-scale graph construction.
Duan et al. (2024)	Proposes HG-CTA, a Heterogeneous Graph-based model for cyber threat attribution, using metapath context-based graph embedding and link prediction.	Performance depends on diverse threat intelligence data; feature extraction is complex and resource-intensive. Model sensitivity to hyperparameters requires careful tuning.
Teuwen (2023)	Presents a modular architecture for cyber threat attribution with a novel Pairing Aggregator for combining outputs from various modules.	Limitations include reliance on simulated data, which may not fully reflect real-world complexities, and challenges in aggregating outputs from individual modules.
Wang et al. (2022)	Models malware behavior using time series shapelets for APT attribution, featuring a novel path generation method and data imbalance handling.	Evaluated on a relatively small dataset; parameter selection for the shapelet model can be challenging.
Zhang et al. (2024)	Introduces a hierarchical attribution framework, leveraging LLMs for automation and practical application in populating threat intelligence.	Dependent on pre-existing attribution reports and heavily reliant on LLMs, potentially limiting comprehensiveness.
Chandra et al. (2023)	Proposes a scalable cybersecurity knowledge graph for APT attribution, integrating real-time data and deep learning to facilitate advanced reasoning for threat detection.	Faces concerns with processing large datasets; relies heavily on expert knowledge for defining reasoning rules.
Edie et al. (2023)	Presents a novel APT attribution approach using association rule mining and a weighted Jaccard similarity index based on MITRE ATT&CK and CTI sources.	Limitations include susceptibility to concept drift as TTPs evolve, and the weighted Jaccard index may not fully capture nuanced relationships.
Rani et al. (2024)	Enhances attribution with ML models, focusing on feature selection and behavioral correlation from malware samples, validated for operational behavior-based attribution.	Reliance on timestamps can be problematic due to tampering; limited dataset for new threat groups.
Lee and Choi (2025)	Introduces the MUCAMP method, leveraging few-shot campaign generation and domain knowledge integration to stress-test classifiers for threat group attribution.	TTP synonym replacement may not capture nuanced behaviors, and high/low mutation levels can distort patterns.
Böge et al. (2025)	Proposes a hybrid deep learning architecture combining transformers and CNNs for cyber threat actor attribution, demonstrating high performance on command sequences.	Reliance solely on command sequences may not provide a comprehensive view of all factors influencing actor behavior.

Together, these recent works (see Table 10) illustrate progress in ML- and AI-driven attribution, yet also reveal gaps—particularly in data completeness, false flag detection, and integration of socio-technical dimensions. Table 11 presents a comparative performance overview, evaluating these approaches using standard metrics such as accuracy, precision, recall, and F1-score. Continued research is needed to develop attribution frameworks that are not only technically robust but also resilient to deception and adaptable to evolving threat landscapes (Skopik and Pahi, 2020).

8. Discussions, implications and future research directions

The escalating sophistication and diversity of cyber threats exemplified by the pervasive use of proxies, false flags, AI-powered attacks, and the temporal compression of attack lifecycles demand a fundamental paradigm shift in attribution methodologies. As this survey has meticulously detailed, attackers are now employing advanced evasion techniques and exploiting vulnerabilities across multiple domains,

Table 11
Quantitative comparison of cyber threat attribution studies.

Research	Attribution scope	Dataset	Tools	Highest accuracy (%) / other metrics	Highest precision (%)	Highest recall (%)	Highest F-measure (%)	FP rate (%)
Noor et al. (2019)	Cyber threat actor	Public CTI data and MITRE ATT&CK (High level IOC)	NLP, Naïve Bayes, KNN, Decision Tree, Random Forest, DLNN	94	92	89	89	2
Puzis and Angappan (2020)	Cyber threat actor	Threat Reports: MITRE ATT&CK	ANN, SIMVER, Word2Vec	86.5	95.4	83.3	87.9	–
Perry et al. (2019)	Cyber threat actor	Public Threat Reports : MITRE ATT&CK, APT groups and operations	NLP, SMOBI, XGBoost	58.4	55	52.4	–	–
Irshad and Siddiqui (2023)	Cyber threat actor	Public Threat Reports from Maultiple sources and Prior Work	Attack2Vec, Random Forest, Decision Tree, SVM	96	96.6	95.58	95.75	–
Basnet et al. (2024)	Cyber threat actor/Threat actor group	Malware samples and reports from Cuckoo Sanbox and VirusTotal, (Behavioral data), Textual threat intelligence reports	Deep Reinforcement Learning	89	87	85	86	–
Mei et al. (2024)	APT attacker	UNSW-NB15	Multi-Layer Perceptual Deep Neural Network	94.56	95.58	94.56	94.94	5.98
Kida and Olukoya (2023)	APT family/group level and the broader nation-state level	Malware samples from a previous Study	Random Forest, SVM, Naïve Bayes, KNN	89	–	–	88	–
Xiao et al. (2024)	APT actor/group	Multisource cyber threat intelligence reports	Heterogeneous attributed graph, BERT, Node2Vec	–	Micro-F1: 83.2	Macro-F1: 70.5	–	–
Irshad and Siddiqui (2024)	Cyber threat actor	Threat Actor Encyclopedia (OSINT by CERT): Technical and Behavioral Features	Decision Tree, Random Forest, SVM, LSTM	97	98.8	97	97.2	–
Ren et al. (2022)	APT groups or organizations	English and Chinese CTI reports	APT knowledge graph, Naïve Bayes, KNN, Decision Tree, Random Forest, SVM, XGBoost, MLP, Graph Neural Network	–	Macro-F1: 73.2 (English), 81.88 (Chinese)	Micro-F1: 73.2 (English), 81.88 (Chinese)	–	–
Vehabovic et al. (2023)	Attributing to Malware Families	Windows 10/11 ransomware	SVM, FNN, XGBoost, RF	94–96 (range)	–	–	–	–
Wang et al. (2021)	APT group	APT malwares from public threat intelligence reports: Code & String Features	RF, DNN	97.87	–	–	–	–
Xiang et al. (2024)	Cyber attacker group or organization	Commercial Real-world network intrusion logs	Word2Vec, K-means and spectral clustering algorithms	88.89	–	–	–	–
Nakhodchi et al. (2021)	Specific type or category of cyber-attack	ICS Cyber-attack Dataset Suite	CatBoost	93–99	93–100	90–98	93–100	–
Jahromi et al. (2021)	Cyber attacker groups or communities	ICS Cyber-attack Dataset Suite	DNN and Decision Tree	98.14, 99.71	98.22, 99.74	98.14, 99.71	98.18, 99.72	–
Leite et al. (2024)	Threat actor	Formulated dataset: annotated DNS records from threat campaigns	DBSCAN and HDBSCAN	73.78	84.38	–	–	–
Duan et al. (2024)	Threat actor/Intrusion set	Public cyber threat intelligence from AlienVault and MITRE ATT&CK	Heterogeneous Graph Embedding Metapath-based context modeling Graph Neural Networks Link Prediction	Mean Rank (MR) - 14.02/Hit@5 - 54.42	–	–	–	–

(continued on next page)

Table 11 (continued).

Teuwen (2023)	Threat actors	Simulated dataset (synthetic Threat actor profiles)	Linear SVMs, Pairing Aggregator, Opinion Pooling Functions (Logarithmic, Linear), XGBoost, Single Linear SVM	81.7 (Highest)				
Wang et al. (2022)	APT groups	APT malwares from public threat intelligence reports (Symantec, FireEye.)	HDBSCAN and RFC	AUC - Micro-average: 0.951, Macro-average: 0.936	92	91	91.5	–
Zhang et al. (2024)	Threat groups/ attackers	Publicly released attribution reports: FireEye, Cisco Talos, ESTsecurity, AhnLab, Telsy	LLM: Alibaba Qwen	N/A	–	–	–	–
Chandra et al. (2023)	APT groups	Threat intelligence feeds, incident reports, social media posts, historical attack data and other open-source repositories	Cybersecurity Knowledge Graph, BI-LSTM with GRU, NLP, Rule-based Reasoning	98	–	–	–	–
Rani et al. (2024)	APT groups	APT malware samples from VirusShare and Malshare and a APT malware Dataset	Naive Bayes,SVM, KNN, Decision Trees, RF ,Logistic Regression, XGBoost, LightGBM, AdaBoost, and Voting classifier.		80.84	74.15	76.55	
Lee and Choi (2025)	Threat groups	Security reports from the APT & Cybercriminals Real world Campaign Collection , Custom-built TTP synonym corpus, Attribution labels from known threat intelligence sources (MITRE)	NLP,Camp2Vec, TF-IDF Embedding, Data Augmentation	Group score: 0.2615	–	–	–	
Böge et al. (2025)	Threat actors/APT groups	Private threat intelligence data from PRODAFT	Hybrid Deep Learning architecture with Transformers and CNN	95.13	–	–	95.11	
Edie et al. (2023)	APT groups	TTP from MITRE ATT&CK and CTI sources	Association Rule Mining, Weighted Jaccard Similarity Index, Activity-Attack Graphs	95.7	–	–	–	–

rendering the task of reliable attribution increasingly challenging (Section 3). The multifaceted nature of cyber threats, underscored by the diverse attribution techniques and challenges discussed, paints a complex picture of the cybersecurity landscape. Our analysis reveals that traditional, deterministic approaches are often insufficient, necessitating a rigorous shift toward data-driven, intelligent, and interdisciplinary methodologies (Section 7-A).

8.1. Strategic implications and policy realities

Accurate and timely cyber threat attribution carries profound implications, extending far beyond the technical realm to directly influence strategic defense, international relations, and legal accountability. It empowers defenders to take decisive action against threat actors, disrupting their operations and deterring future attacks. Attribution also serves as a critical foundation for international cooperation, legal action, and policy decisions aimed at combating cybercrime on a global scale. Implications for Key Stakeholders include:

For Intelligence Communities (IC): The IC must shift from a reactive “who” to a proactive “why”, prioritizing motive analysis over static IOC matching. This requires embracing AI-human hybrid workflows, where advanced AI (e.g., GNNs for infrastructure mapping) augments human analysts’ geopolitical intent assessment.

For Legal Systems: Legal frameworks need to adapt to digital evidence. This involves standardizing Cyber Forensic Chain-of-Custody protocols to ensure the admissibility of technical evidence in international tribunals, aligning with evolving regulations.

For the Private Sector: Organizations must move beyond mere compliance to strategic investment. This includes developing Attribution ROI Models to calculate the cost of inaction (e.g., reputation damage from non-disclosure) versus the cost of errors (e.g., stock dips after false attribution). Investing in secure attribution sharing platforms and internal expertise is crucial.

For Global Governance: There is an imperative to establish clear international norms for cyber behavior. This includes implementing Attribution Confidence Tiers (e.g., UN-backed frameworks for diplomatic responses ranging from sanctions for definitive attribution to joint investigations for ambiguous cases) to formalize accountability and guide calibrated international responses.

Policy Realities and Trade-offs: The decision to attribute publicly is often a complex geopolitical calculus, weighing the benefits of deterrence against potential diplomatic repercussions or escalation. There is a continuous tension between public attribution and covert response options. Public naming can act as a strong deterrent, but it may also compromise sensitive intelligence sources or provoke further adversarial adaptation (Deibert, 2017). This leads to alternative paradigms such

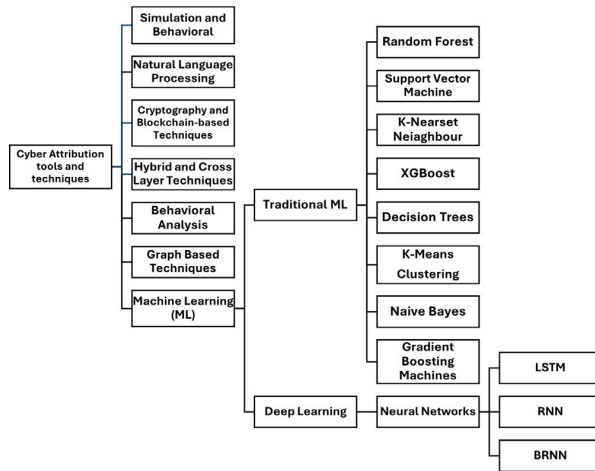


Fig. 9. Taxonomy of emerging approaches in cyber threat attribution.

as “attributionless defense”, which prioritizes resilience and disruption without necessarily naming the perpetrator. While such approaches are valuable for immediate defense, they may fall short in achieving long-term deterrence or accountability.

Furthermore, implementing advanced attribution capabilities necessitates a careful Cost-Benefit Analysis. The ROI of deploying sophisticated technologies like Heterogeneous Graph Neural Networks (HGNNs) versus simpler clustering methods must consider not only detection accuracy but also the significant computational, resource, and human capital investments. This directly links to the Resource and Expertise Challenges (Section 3), where the disparity between large and small organizations impacts the feasibility of adopting cutting-edge solutions.

The legal admissibility of AI-driven or blockchain-verified attributions remains a critical concern. While AI-powered analytics can enhance evidence collection, the “black-box” nature of some deep learning models presents challenges for interpretability in court, and there is a current lack of established legal precedents for such evidence. Similarly, while blockchain offers immutability for evidence, its legal standing for cross-border cyber forensics requires further research and harmonization of international legal frameworks. This underscores the imperative for continuous dialogue between cybersecurity experts, legal professionals, and policymakers.

Finally, establishing an Attribution Maturity Model is crucial for organizations and nations to assess and advance their capabilities systematically. Such a staged framework (e.g., progressing from basic IOC Matching to advanced Predictive Actor Profiling) would provide a roadmap for developing more robust, proactive attribution strategies, guiding investment and capability development.

8.2. Future research imperatives

Addressing the complexities outlined throughout this survey demands a concerted and multi-pronged approach to future research, focusing on both methodological innovations and policy-oriented solutions.

8.2.1. Methodological innovations needed

Future research must prioritize the development of more sophisticated AI/ML models capable of handling the complexity, dynamism, and adversarial nature of cyberattacks. This includes advancing techniques to mitigate algorithmic bias and enhance transparency (Explainable AI - XAI) (Hassija et al., 2024), fostering trust and enabling human

analysts to validate AI-driven decisions. The exploration of privacy-preserving attribution techniques, such as federated learning, will be essential for enabling collaborative data sharing across organizations and jurisdictions without compromising sensitive information (Amara et al., 2024; Sikos, 2020; Jayasingh et al., 2016). Research into adversarial robustness of attribution models is also vital, ensuring their resilience against deliberate attempts at deception and manipulation (Zhu et al., 2024; Dixit, 2024).

Beyond AI, the continued development and integration of cutting-edge technologies are paramount. Quantum-resistant forensic hashing is required to secure the integrity of AI-generated artifacts against future quantum threats. Game-theoretic models are essential for anticipating adversarial attribution strategies and developing more resilient defensive tactics. Blockchain technology holds immense promise for ensuring the integrity and immutability of forensic evidence, providing a verifiable chain of custody for digital artifacts (Chatziamanetoglou and Rantos, 2024; Gunawardena, 2023; Glady et al., 2024). Advancements in quantum computing could revolutionize the processing speed and analytical capacity for massive datasets, enabling near real-time attribution. Furthermore, embedding ML algorithms within Internet of Things (IoT) devices could enable decentralized threat detection and attribution at the edge, creating more agile response mechanisms.

8.2.2. Policy recommendations for research

Policy-oriented research is crucial to translate technical advancements into effective governance. This includes establishing standardized evaluation benchmarks for new attribution systems (similar to NIST-style tests), ensuring consistency and reliability across different models and methodologies (Yang et al., 2009; Alghamdi, 2021). Research should also focus on developing and implementing cross-border attribution treaties, modeled on international cooperation initiatives but enhanced with ML-driven evidence validation mechanisms. Designing comprehensive legal frameworks for data sharing and evidence admissibility in the AI era, including international arbitration mechanisms, is vital for fostering effective collaboration and prosecution. Finally, a deeper understanding of the ethical and legal implications of automated attribution, particularly concerning potential biases and accountability frameworks, will be crucial in navigating the complex interplay between technology, security, and human rights.

9. Conclusion

Conclusion The cyber threat landscape continues its rapid evolution, driven by increasingly sophisticated adversaries leveraging advanced technologies and complex obfuscation tactics. This dynamic environment renders traditional attribution methodologies insufficient, underscoring the urgent need for a paradigm shift toward data-driven, intelligent, and interdisciplinary approaches. This survey has critically evaluated the multifaceted challenges in attribution, analyzed the capabilities and limitations of existing frameworks, and explored the transformative potential of emerging technologies such as AI/ML, graph analysis, and blockchain. By providing a novel taxonomy and offering actionable insights derived from a rigorous systematic review, this paper aims to equip researchers, practitioners, and policymakers with a structured framework to advance attribution accuracy and resilience. The future of cyber threat attribution lies in the convergence of diverse research avenues, driven by a commitment to accuracy, transparency, and ethical responsibility. As the cyber threat landscape continues to evolve, the quest for effective attribution remains a critical endeavor, shaping the future of cybersecurity and international cooperation in the digital age.

CRedit authorship contribution statement

Nilantha Prasad: Writing – original draft, Validation, Formal analysis, Conceptualization. **Abebe Diro:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **Matthew Warren:** Validation, Supervision. **Mahesh Fernando:** Validation, Supervision, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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